

Chapter 3: Biological Molecules

- What kind of molecules make up living bodies?
- How do we build & break down (digest) those molecules?
- What are the four classes of molecules that build our bodies & that we have to eat?

Nov. 10, 2025

Corresponds with OpenStax Biology 2e Chapter 3

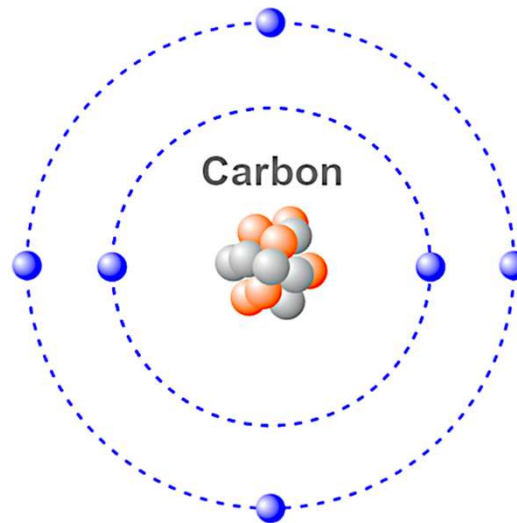
Chapter 3: Biological Molecules

What class of molecules make up living bodies?

- Living things are made up of **organic molecules**
 - Plants make their own organic molecules, plus gain some through their roots
 - Animals eat organic molecules to build & fuel their bodies
- **Organic molecules** = all have carbon & hydrogen
 - Must have both – if a molecule only has 1 of those elements (or neither), it is inorganic
 - Organic molecules contain other elements too

- Why are all organic molecules (& thus all living things) carbon based?
 - **Carbon** = small & versatile: can make 4 different bonds with many different atoms
 - *at least 1 must be hydrogen to be organic*

THE VERSATILITY OF CARBON



Carbon can hold 4 more electrons in the second shell, so can make 4 bonds

- Organic molecules have functional groups

– **Functional group** = small groups of non-hydrogen atoms in organic molecules

- they make each organic molecule unique: protein vs. carbohydrate vs. fat...*

Each functional group gives a molecule a unique property

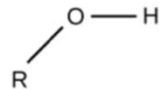
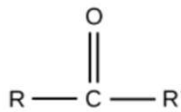
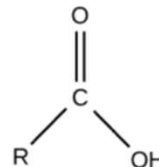
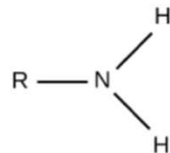
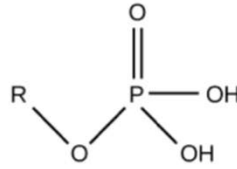
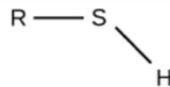
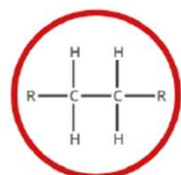
Functional Group	Structure	Properties
Hydroxyl		Polar
Methyl	$R - CH_3$	Nonpolar
Carbonyl		Polar
Carboxyl		Charged, ionizes to release H^+ . Since carboxyl groups can release H^+ ions into solution, they are considered acidic.
Amino		Charged, accepts H^+ to form NH_3^+ . Since amino groups can remove H^+ from solution, they are considered basic.
Phosphate		Charged, ionizes to release H^+ . Since phosphate groups can release H^+ ions into solution, they are considered acidic.
Sulfhydryl		Polar

Figure from OpenStax Biology 2e. *R*, also known as *R*-group, is an abbreviation for any group in which a carbon or hydrogen atom is attached to the rest of the molecule.

Functional Groups in Organic Chemistry

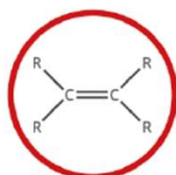
FUNCTIONAL GROUPS ARE GROUPS OF ATOMS IN ORGANIC MOLECULES THAT ARE RESPONSIBLE FOR THE CHARACTERISTIC CHEMICAL REACTIONS OF THOSE MOLECULES. IN THE GENERAL FORMULAE BELOW, 'R' REPRESENTS A HYDROCARBON GROUP OR HYDROGEN, AND 'X' REPRESENTS ANY HALOGEN ATOM.

● HYDROCARBONS
 ● SIMPLE OXYGEN HETEROATOMICS
 ● HALOGEN HETEROATOMICS
 ● CARBONYL COMPOUNDS
 ● NITROGEN BASED
 ● SULFUR BASED
 ● AROMATIC



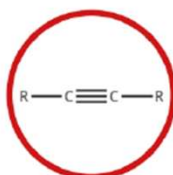
ALKANE

Naming: -ane
e.g. ethane



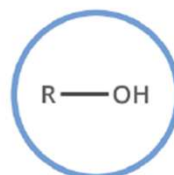
ALKENE

Naming: -ene
e.g. ethene



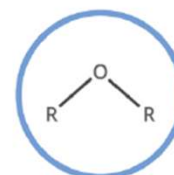
ALKYNE

Naming: -yne
e.g. ethyne



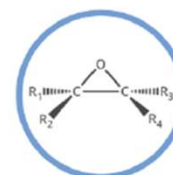
ALCOHOL

Naming: -ol
e.g. ethanol



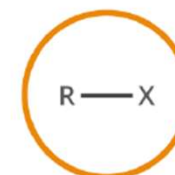
ETHER

Naming: -oxy -ane
e.g. methoxyethane



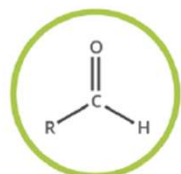
EPOXIDE

Naming: -ene oxide
e.g. ethene oxide



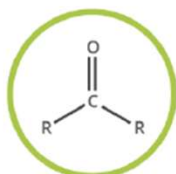
HALOALKANE

Naming: halo-
e.g. chloroethane



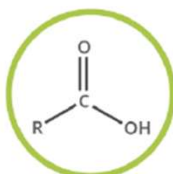
ALDEHYDE

Naming: -al
e.g. ethanal



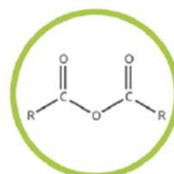
KETONE

Naming: -one
e.g. propanone



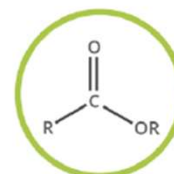
CARBOXYLIC ACID

Naming: -oic acid
e.g. ethanoic acid



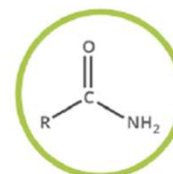
ACID ANHYDRIDE

Naming: -oic anhydride
e.g. ethanoic anhydride



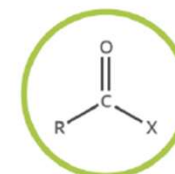
ESTER

Naming: -yl -oate
e.g. ethyl ethanoate



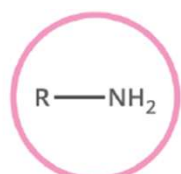
AMIDE

Naming: -amide
e.g. ethanamide



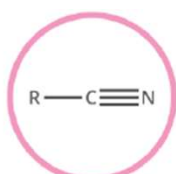
ACYL HALIDE

Naming: -oyl halide
e.g. ethanoyl chloride



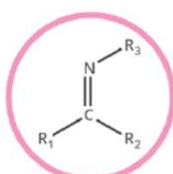
AMINE

Naming: -amine
e.g. ethanamine



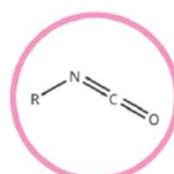
NITRILE

Naming: -nitrile
e.g. ethanenitrile



IMINE

Naming: -imine
e.g. ethanimine



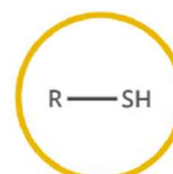
ISOCYANATE

Naming: -yl isocyanate
e.g. ethyl isocyanate



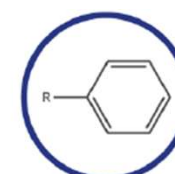
AZO COMPOUND

Naming: -azo-
e.g. azoethane



THIOL

Naming: -thiol
e.g. methanethiol



ARENE

Naming: -yl benzene
e.g. ethyl benzene

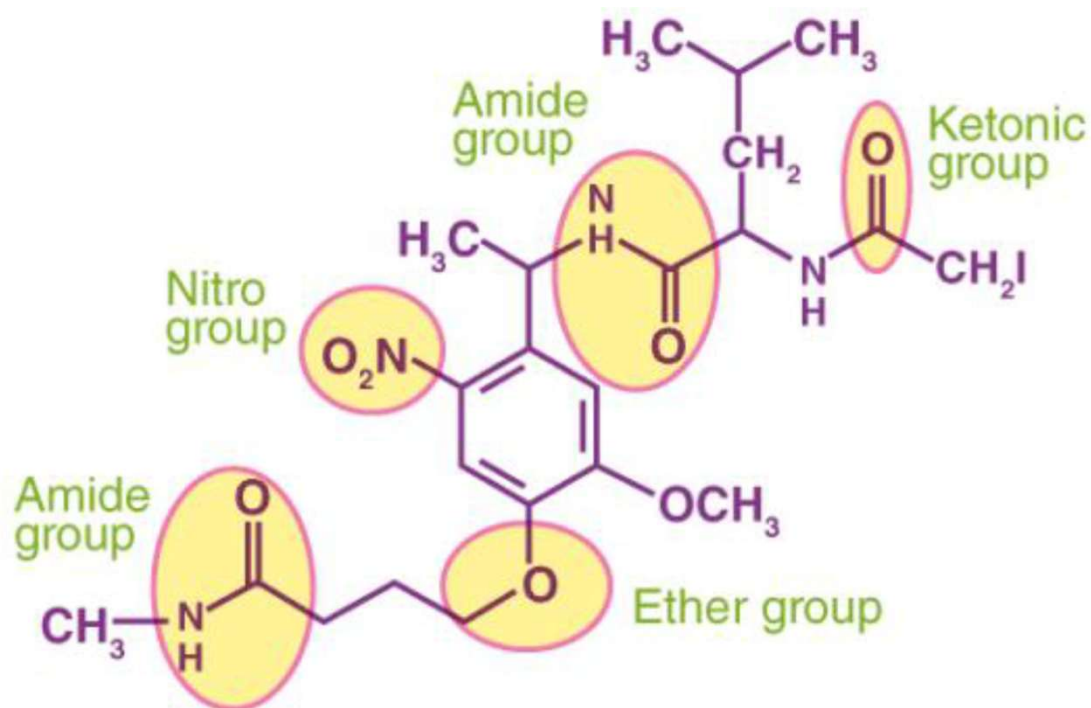


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Functional Groups in Organic Chemistry

Functional groups are the moieties which exhibit their own **distinct features** and properties independent of the molecule they are attached to.

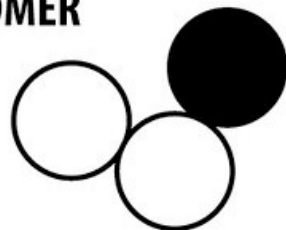


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How do we build & break down (digest) organic molecules?

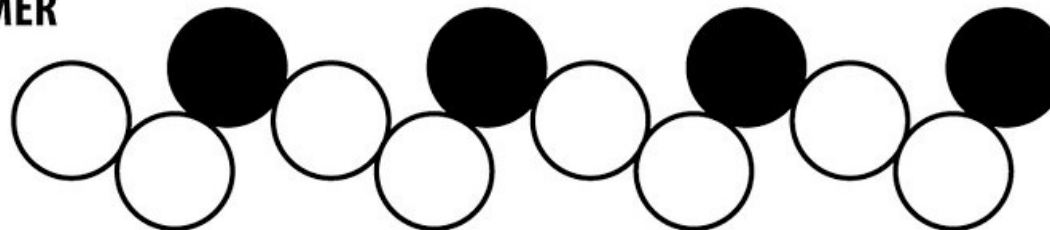
- Small organic molecules = **monomers**
- Monomers are joined to form bigger organic molecules = **polymers**

MONOMER



A monomer is a small molecule.

POLYMER

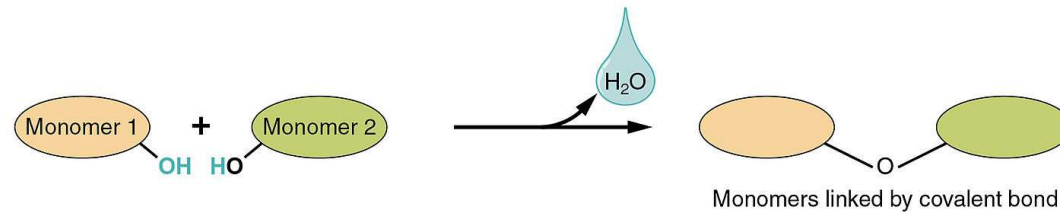


A polymer is a long-chain molecule made up of a repeated pattern of monomers.

- Monomers are linked together into polymers through **dehydration synthesis**
 - e.g. building muscle & other complex molecules

(a) Dehydration synthesis

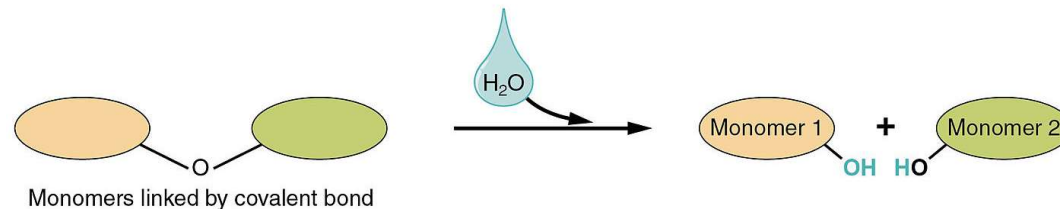
Monomers are joined by removal of OH from one monomer and removal of H from the other at the site of bond formation.



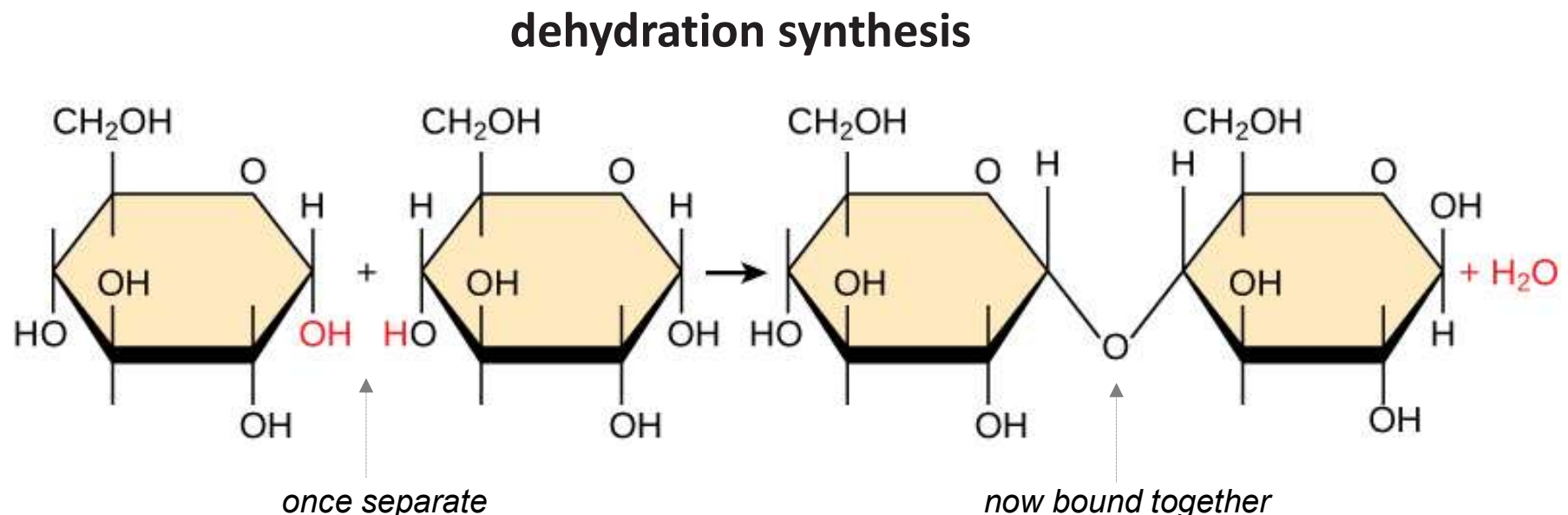
- Polymers are broken down into monomers through **hydrolysis**
 - e.g. digestion & breakdown of molecules for energy

(b) Hydrolysis

Monomers are released by the addition of a water molecule, adding OH to one monomer and H to the other.

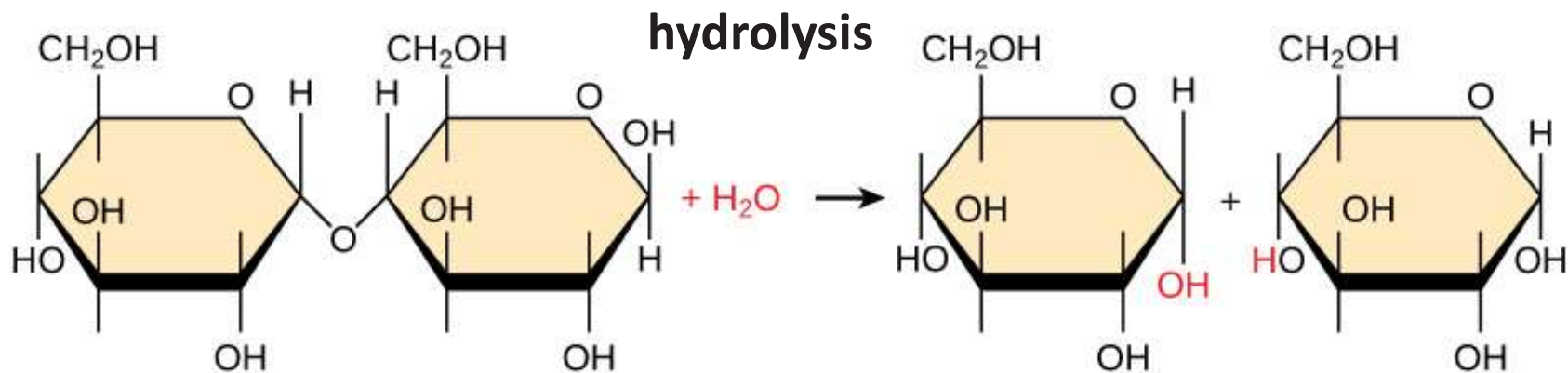


- In **dehydration synthesis**:
 - H & OH removed = loss of water molecule (H_2O)
 - Loss of H_2O means atoms want to bond to stabilize (by forming covalent bonds)
 - Bond together: monomer + monomer = polymer



- In **hydrolysis**:

- Add water, which is separated into H & OH
- H & OH are inserted, breaking the bond between monomers
 - Digestion occurs: polymer broken down into individual monomers



What are the four classes of molecules that build our bodies & that we have to eat?

- Organic molecules fall into 4 main categories
 - **Carbohydrates**: our main energy source & some for structure
 - **Proteins**: do nearly everything that keeps us alive
 - **Nucleic Acids**: energy carriers, cell messengers, & hereditary information
 - **Lipids**: energy storage, waterproofing, cholesterol, hormones, & cell membranes

Macromolecules

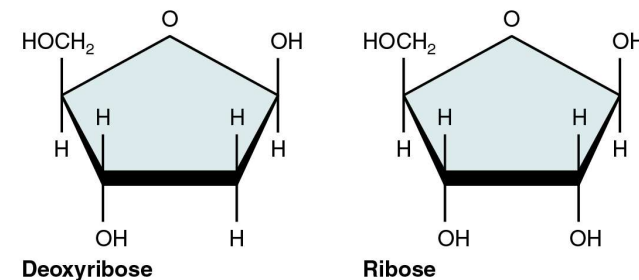
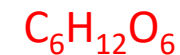
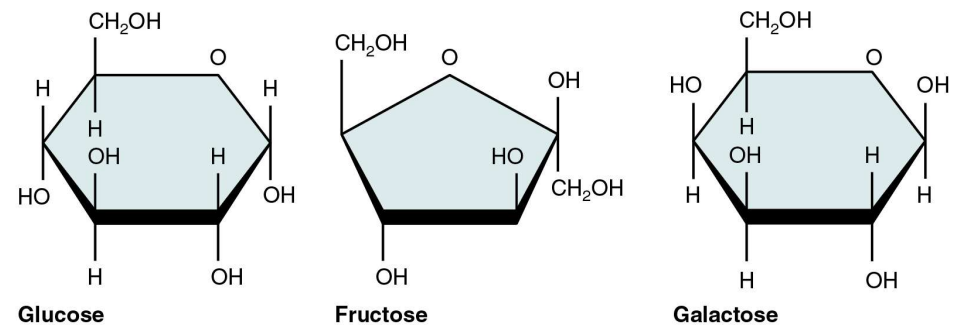
Macromolecule	Subunit	Function	Example
PROTEINS			
Globular	Amino acids	Modified glucose	Hemoglobin
Structural	Amino acids	Catalysis; transport Support	Hair; silk
NUCLEIC ACIDS			
DNA	Nucleotides	Encodes genes	Chromosomes
RNA	Nucleotides	Needed for gene expression	Messenger RNA
LIPIDS			
Fats	Glycerol and three fatty acids	Energy storage	Butter; corn oil; soap
Phospholipids	Glycerol, two fatty acids, phosphate, and polar R groups	Cell membranes	Lecithin
Prostaglandins	Five-carbon rings with two nonpolar tails	Chemical messengers	Prostaglandin E (PGE)
Steroids	Four fused carbon rings	Membranes; hormones	Cholesterol; estrogen
Terpenes	Long carbon chains	Pigments; structural	Carotene; rubber
CARBOHYDRATES			
Starch, glycogen	Glucose	Energy storage	Potatoes
Cellulose	Glucose	Cell walls	Paper; strings of celery
Chitin		Structural support	Crab shells

- **Carbohydrates** = sugars, hydrophilic
 - 3 types based on size: monosaccharides, disaccharides, polysaccharides

– **Monosaccharide** = a carb of just 1 sugar molecule

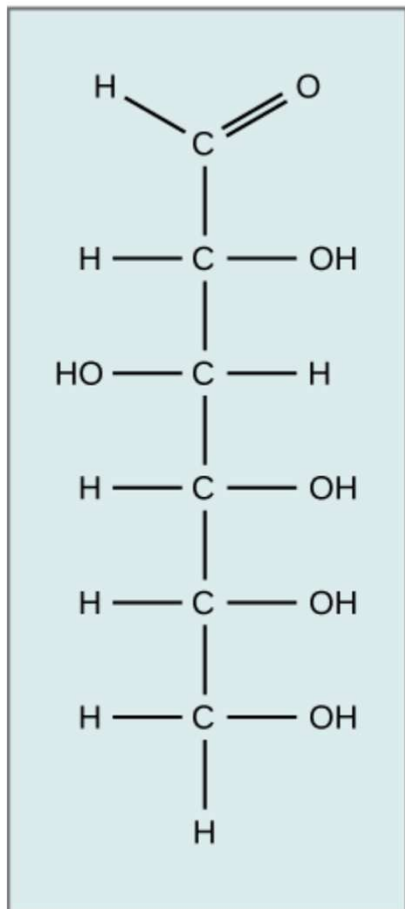
- “simple sugar”
- sweet-tasting on the tongue
- *monomer*

Monosaccharide examples
(glucose is the most common)

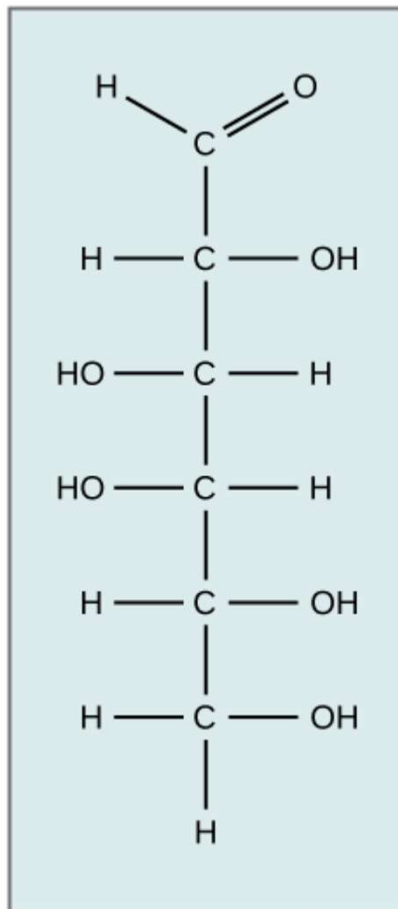


Monosaccharide Examples

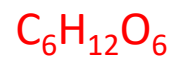
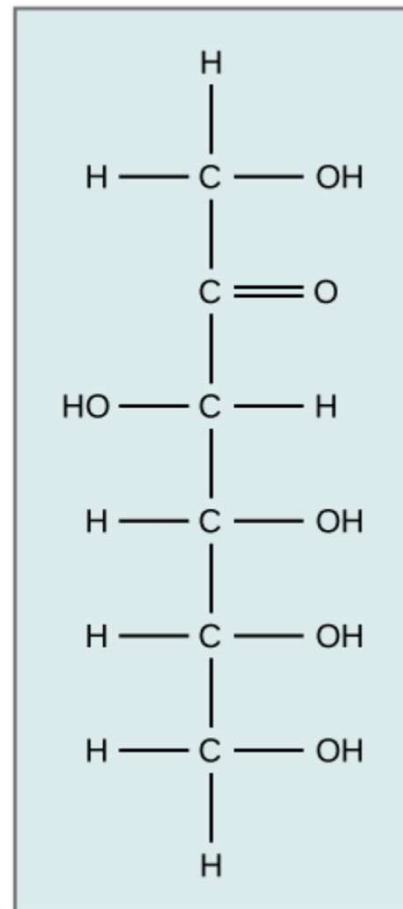
Glucose



Galactose



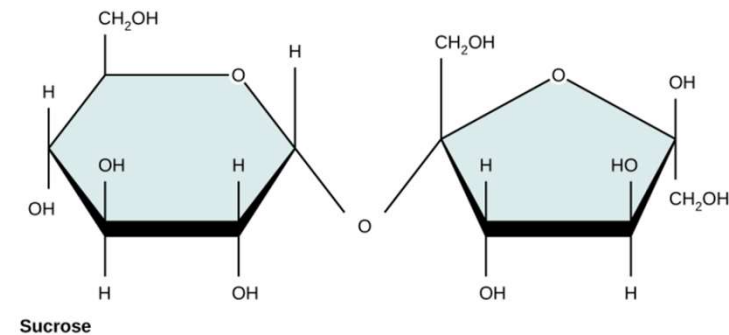
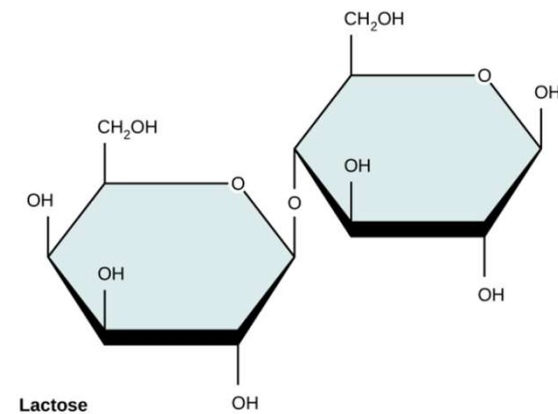
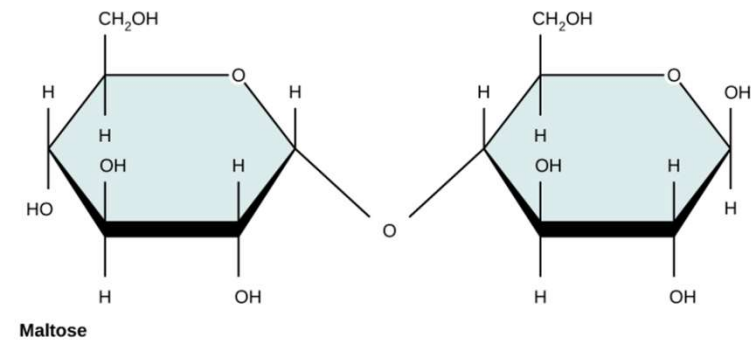
Fructose



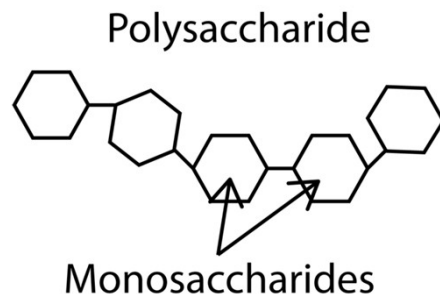
– **Disaccharide** = 2 monosaccharides linked together

- also sweet (table sugar)
- *small polymer*

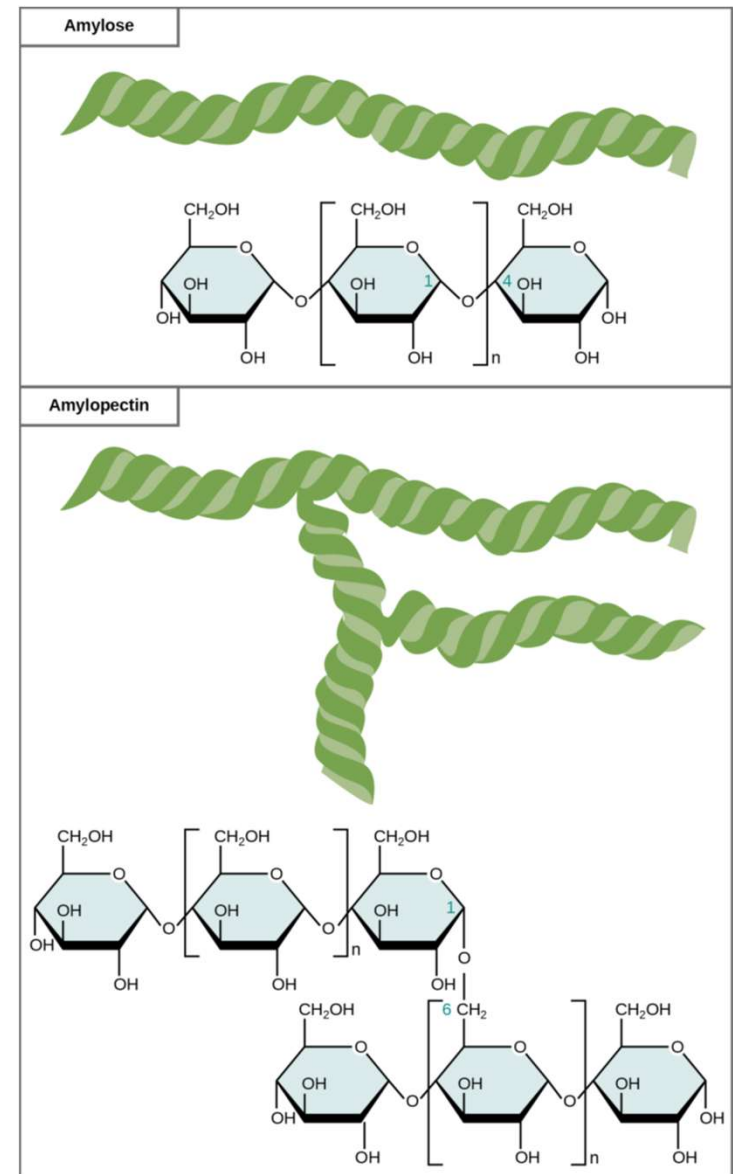
disaccharide examples



- **Polysaccharide** = a *polymer* of many monosaccharides linked together
 - “complex carb”
 - NOT sweet-tasting



Starch, glycogen, cellulose, and chitin are primary examples of polysaccharides.

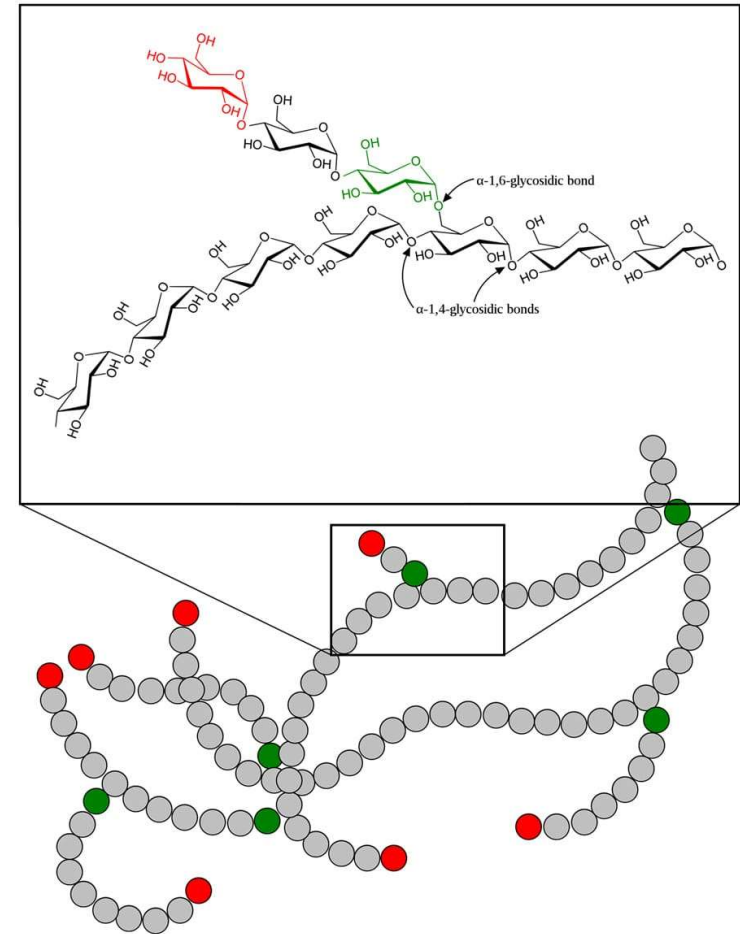


Amylose and amylopectin are two different starch forms

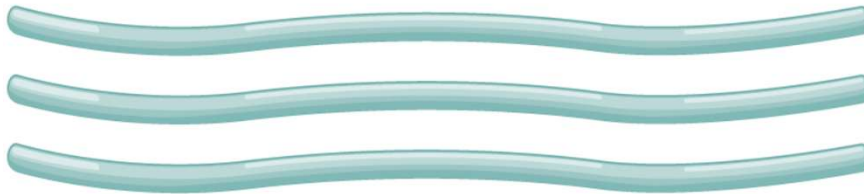
Polysaccharide Examples

Glycogen is the storage form of glucose in humans and other vertebrates

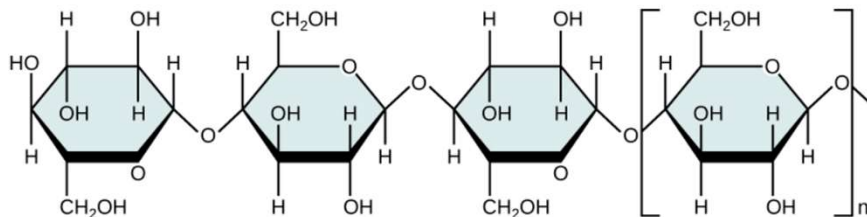
Whenever blood glucose levels decrease, glycogen breaks down to release glucose in a process scientists call glycogenolysis.



Cellulose fibers



Cellulose structure

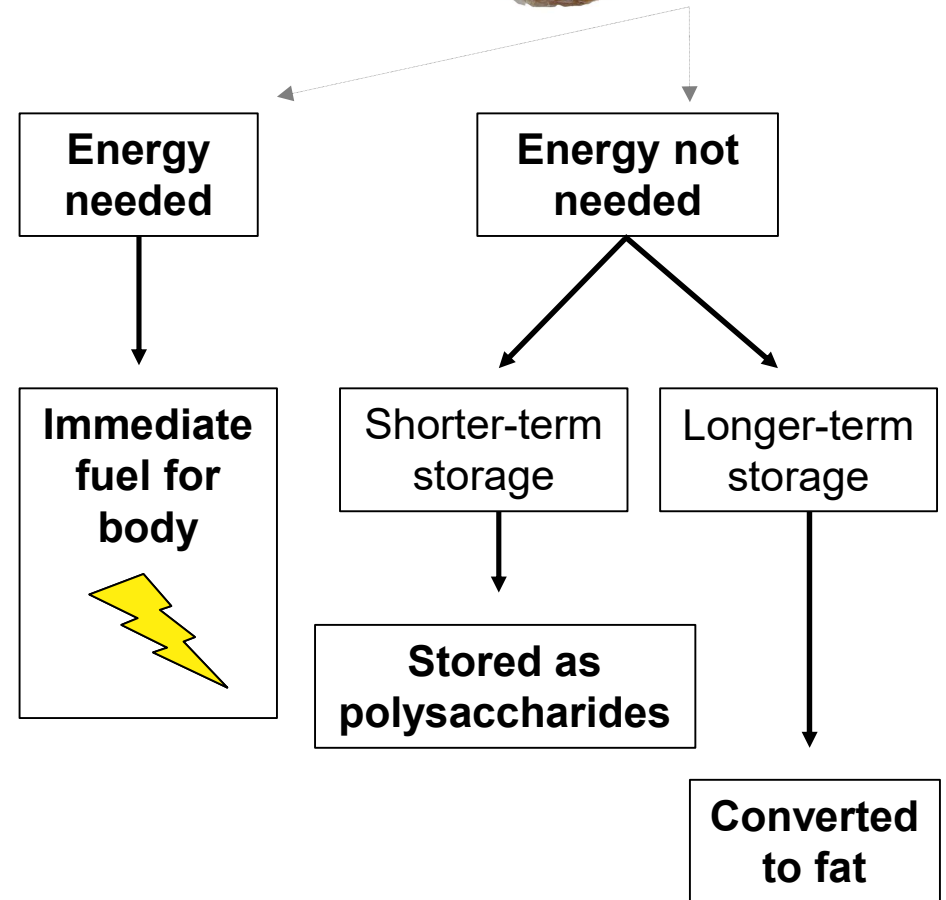


Cellulose is the most abundant natural biopolymer

- In the body, **monosaccharides** are:

- used for immediate energy

- or joined together (dehydration synthesis) as polysaccharides or fats & stored for later energy use



- In the body, **disaccharides** are:
 - used for short-term energy storage
 - when energy is needed, they are broken apart into monosaccharides (by hydrolysis)
 - sucrose = table sugar, the most common disaccharide

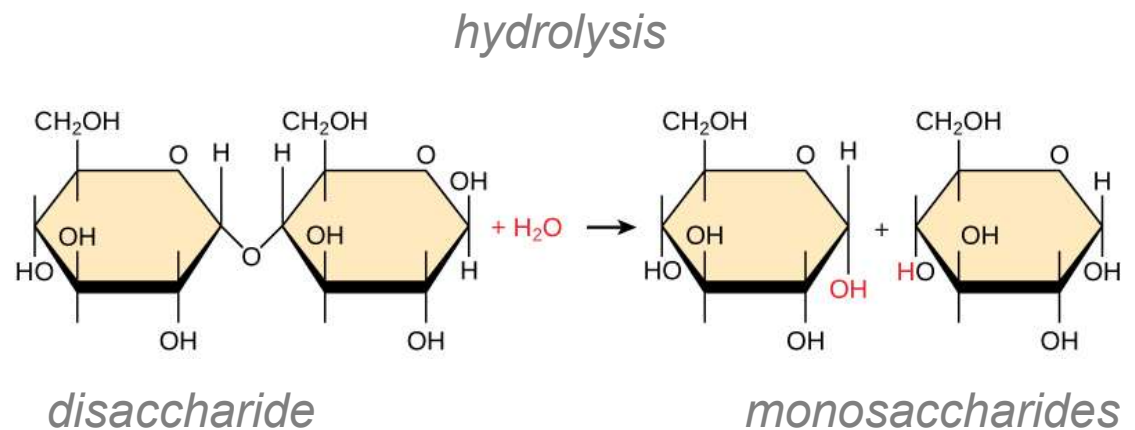
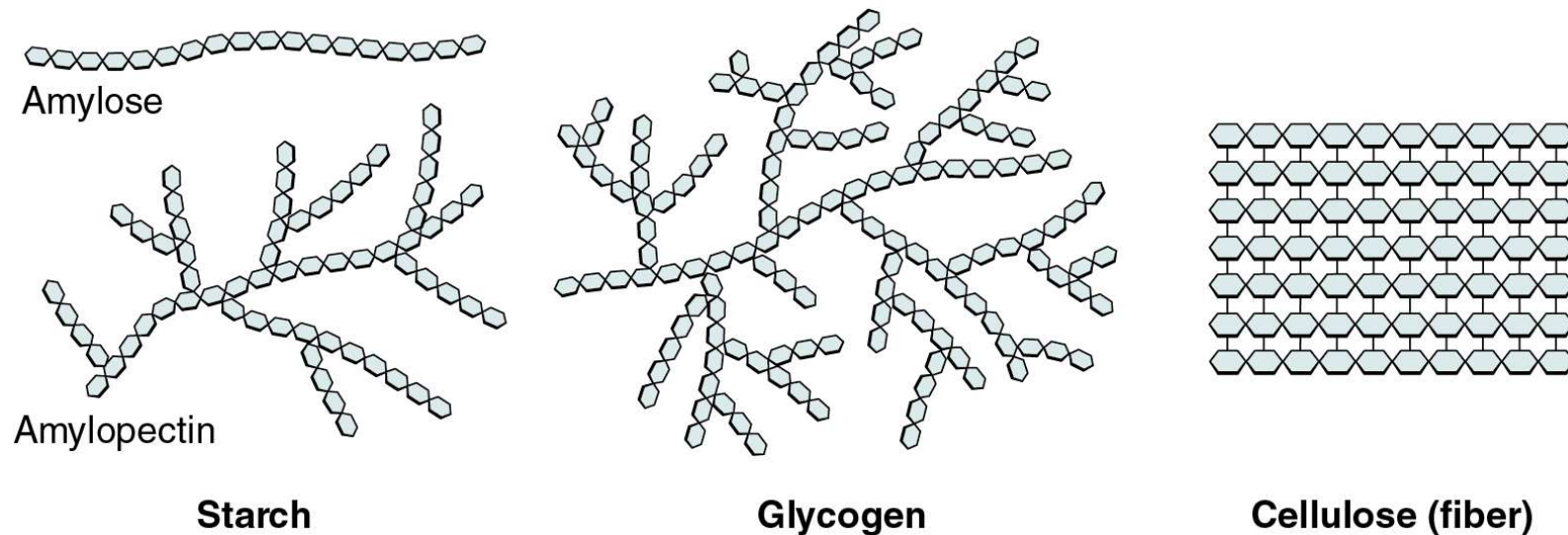


Figure: https://world.openfoodfacts.org/cgi/product_image.pl?code=4864394070294&id=front_ka

Figure: https://commons.wikimedia.org/wiki/File:Figure_03_01_02.jpg

- In the body, **polysaccharides** are used for:
 - long-term energy storage
 - starch (in plants) & glycogen (in animals)
 - structural support
 - e.g. cellulose (fiber) in plants



<http://openstax.org/l/carbohydrates>

- Carbohydrates (sugars) & blood sugar levels
 - Monosaccharides & disaccharides lead to quick, short increases in blood sugar
 - *if a lot of sugar is consumed, it may even crash afterward, leading to fatigue, headaches, hunger, etc.*
 - Polysaccharides lead to slow, longer-lasting increases or maintenance

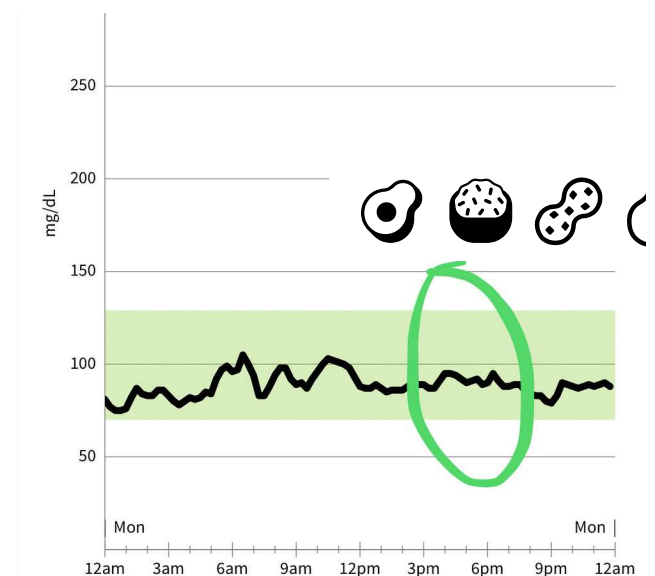
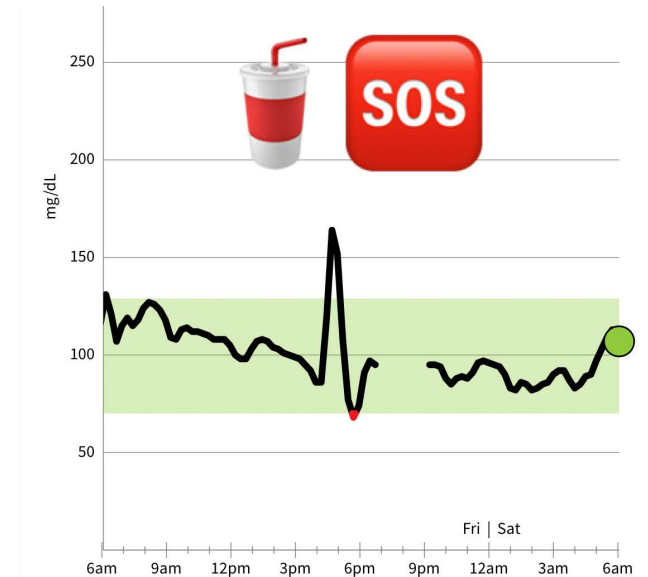


Figure: <https://www.tedeytan.com/2019/05/13/33492>

Figure: <https://www.tedeytan.com/2019/04/28/32835>

- Carbohydrates on nutrition labels
 - “Sugars” = monosaccharides & disaccharides
 - “Dietary Fiber” = not digested carbs
 - helps with movement through digestive system
 - helps regulate blood sugar
 - Rest of “Total Carbohydrates” = polysaccharides

Nutrition Facts	
3 servings per container	
Serving size	(75g)
Amount Per Serving	
Calories	20
% Daily Value*	
Total Fat 0g	0%
Saturated Fat 0g	0%
Trans Fat 0g	
Sodium 140mg	6%
Total Carbohydrate 4g	1%
Dietary Fiber 2g	7%
Total Sugars 0g	
Includes 0g Added Sugars	0%
Protein 2g	4%
Not a significant source of cholesterol, vitamin D, calcium, iron, and potassium	
*The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.	

- Fructose = a monosaccharide
 - **High fructose corn syrup** = high amounts of fructose & some glucose
 - made in a manufacturing plant
 - has replaced table sugar in many food & drinks
 - cheaper than sugar

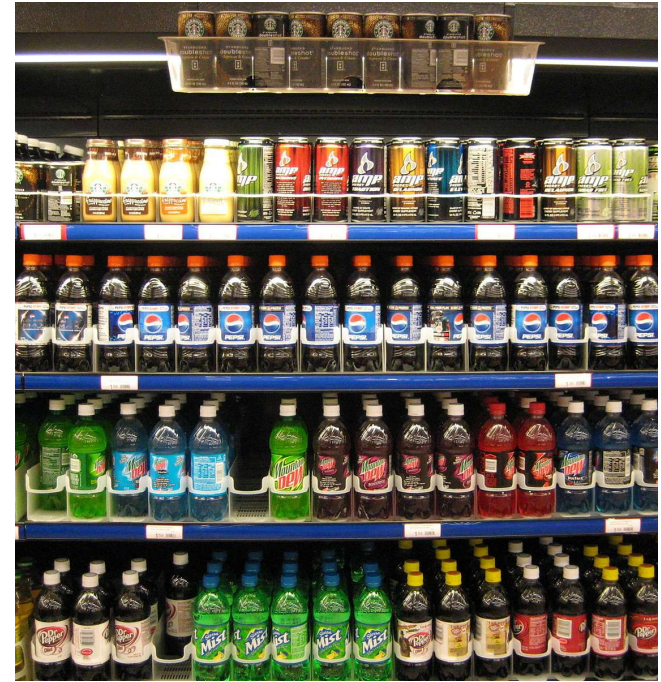


Figure: <https://commons.wikimedia.org/wiki/File:Sodas.JPG>

Figure: <https://www.flickr.com/photos/fanofretail/35636132533>

- High Fructose Corn Syrup & health – dangers?
 - fructose is immediately stored as fat in the liver (instead of used as energy or stored as fat elsewhere in the body)
 - likely linked to:
 - diabetes & other metabolic disorders
 - fatty liver disease
 - Inflammation
 - best to avoid or consume only in moderation



Figure: <https://www.flickr.com/photos/oregonstateuniversity/8167345874>

Figure: <https://www.pxfuel.com/en/free-photo-expxe>

- **Artificial sweeteners** = disaccharides similar to table sugar, but are non-digestible: have no calories
 - Found in diet, sugar-free, & low-carb food & drinks
 - Common artificial sweeteners:
 - Sucralose: e.g. “Splenda”
 - Saccharin: e.g. “Sweet ’N Low”
 - Aspartame: e.g. “Equal”



- Artificial sweeteners & health issues
 - It's hard to know how safe the traditional artificial sweeteners are:
 - many studies show they are safe, but others have found links to insulin resistance/ diabetes, weight gain, migraines, changes in gut health, & other issues

Screenshot = pubmed.com search for review articles covering "artificial sweeteners"

1 [The Association Between Artificial Sweeteners and Obesity.](#)
 Pearlman M, Obert J, Casey L.
 Curr Gastroenterol Rep. 2017 Nov 21;19(12):64. doi: 10.1007/s11894-017-0602-9. PMID: 29159583 Review.
 RECENT FINDINGS: Although **artificial sweeteners** were developed as a sugar substitute to help reduce insulin resistance and obesity, data in both animal models and humans suggest that the effects of **artificial sweeteners** may contribute to metabolic synd ...

2 [Non-caloric artificial sweeteners and the microbiome: findings and challenges.](#)
 Suez J, Korem T, Zilberman-Schapira G, Segal E, Elinav E.
 Gut Microbes. 2015;6(2):149-55. doi: 10.1080/19490976.2015.1017700. Epub 2015 Apr 1. PMID: 25831243 [Free PMC article.](#) Review.
 Non-caloric **artificial sweeteners** (NAS) are common food supplements consumed by millions worldwide as means of combating weight gain and diabetes, by retaining sweet taste without increasing caloric intake. ...

3 [Artificial sweeteners: a systematic review of metabolic effects in youth.](#)
 Brown RJ, de Banate MA, Rother KI.
 Int J Pediatr Obes. 2010 Aug;5(4):305-12. doi: 10.3109/17477160903497027. PMID: 20078374 [Free PMC article.](#) Review.
 However, recent animal studies provide intriguing information that supports an active metabolic role of **artificial sweeteners**. This systematic review examines the current literature on **artificial** sweetener consumption in children and its health effects. ...Ra ...

4 [The role of artificial and natural sweeteners in reducing the consumption of table sugar: A narrative review.](#)
 Mooradian AD, Smith M, Tokuda M.
 Clin Nutr ESPEN. 2017 Apr;18:1-8. doi: 10.1016/j.clnesp.2017.01.004. Epub 2017 Feb 4. PMID: 29132732 Review.
 Commonly used non-caloric **artificial sweeteners** may have unfavorable effect on health including glucose intolerance and failure to cause weight reduction. The nutritive **sweeteners** include sugar alcohols such as sorbitol, xylitol, lactitol, mannitol, erythritol ...

5 [Artificial sweeteners: safe or unsafe?](#)
 Qurrat-ul-Ain, Khan SA.
 J Pak Med Assoc. 2015 Feb;65(2):225-7. PMID: 25842566 Review.
Artificial sweeteners or intense **sweeteners** are sugar substitutes that are used as an alternative to table sugar. ...There is some ongoing debate over whether **artificial** sweetener usage poses a health threat. This review article aims to cover the health ...

6 [Artificial sweeteners--do they bear a carcinogenic risk?](#)
 Weihrauch MR, Diehl V.
 Ann Oncol. 2004 Oct;15(10):1460-5. doi: 10.1093/annonc/mdh256. PMID: 15367404 [Free article.](#) Review.
Artificial sweeteners are added to a wide variety of food, drinks, drugs and hygiene products. ...However, according to the current literature, the possible risk of **artificial sweeteners** to ...

- Some of the newer artificial sweeteners are more naturally derived & seem to have better health effects
 - Monk fruit: e.g. nectresse
 - Sugar alcohols: e.g. sorbitol, xylitol, & erythritol
 - Stevia: also called RebA, Truvia, PureVia, etc.



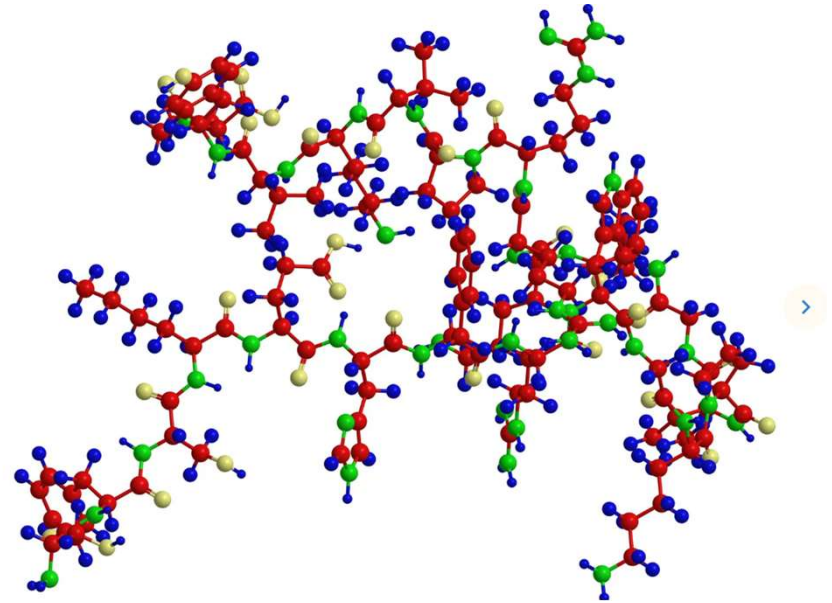
Figure: <https://www.flickr.com/photos/jeepersmedia/14977306178>

Figure: <https://world.openfoodfacts.org/product/6600426740982/erythritol-sugar-replacement-buy-trusted-food>

Figure: <https://www.flickr.com/photos/jeepersmedia/16384420251>

■ Proteins

- A protein is a naturally occurring, extremely complex substance that consists of amino acid residues joined by peptide bonds.
- Proteins are present in all living organisms and include many essential biological compounds such as enzymes, hormones, and antibodies.
- Proteins are of great nutritional value and are directly involved in the chemical processes essential for life.



The molecular structure of a peptide (a small protein) consists of a sequence of amino acids.



Beans, lentils, peas, and other legumes are high in protein and contain many essential amino acids.

■ Proteins

- The importance of proteins was recognized by chemists in the early 19th century, including Swedish chemist Jöns Jacob Berzelius, who in 1838 coined the term protein, a word derived from the Greek *prōteios*, meaning “holding first place.”
- Proteins are species-specific; that is, the proteins of one species differ from those of another species. They are also organ-specific; for instance, within a single organism, muscle proteins differ from those of the brain and liver.

- **Proteins** do nearly everything that make us “us” & keep us alive
 - Human bodies make over 10,000 different proteins
 - a problem with only ONE can = illness or death

Protein Types and Functions

Type	Examples	Functions
Digestive Enzymes	Amylase, lipase, pepsin, trypsin	Help in food by catabolizing nutrients into monomeric units
Transport	Hemoglobin, albumin	Carry substances in the blood or lymph throughout the body
Structural	Actin, tubulin, keratin	Construct different structures, like the cytoskeleton
Hormones	Insulin, thyroxine	Coordinate different body systems' activity
Defense	Immunoglobulins	Protect the body from foreign pathogens
Contractile	Actin, myosin	Effect muscle contraction
Storage	Legume storage proteins, egg white (albumin)	Provide nourishment in early embryo development and the seedling

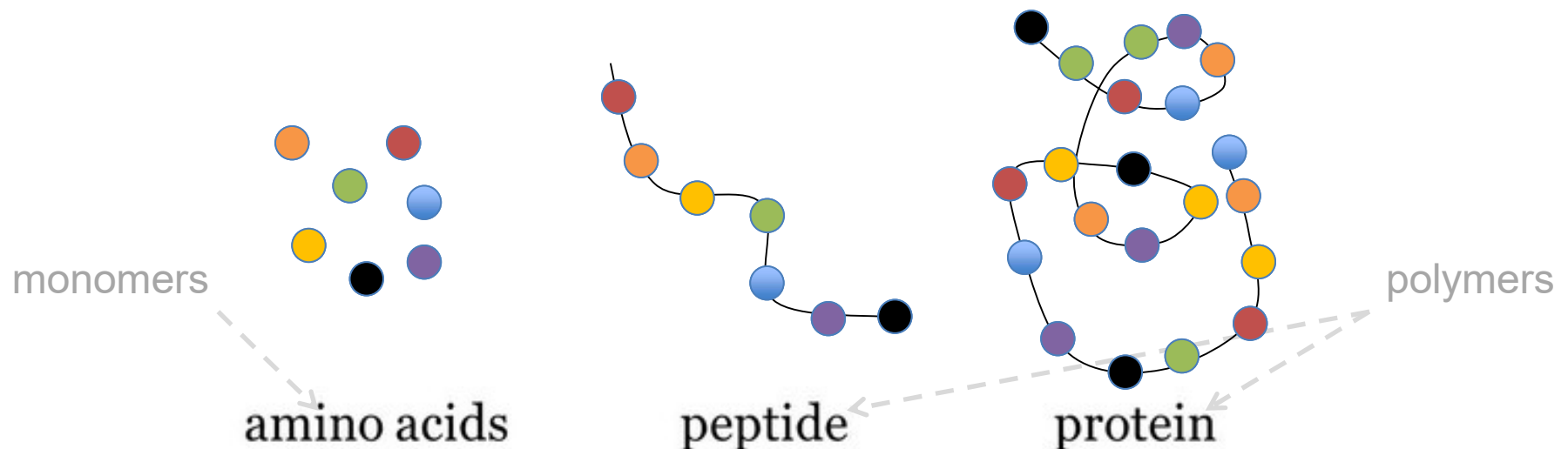
These are just a few types of proteins & their functions

■ Proteins have so many functions

- Make up muscles
- Act as hormones
- Regulate blood sugar
- Carry oxygen to all cells
- Allow sperm cells to swim
- Significant component of bone
- Clot blood when wounds occur
- Make up skin, feathers, horns, & hair
- Make up the pigments that define eye, hair, & skin color
- Act as enzymes, assisting chemical reactions in our bodies
- Make up immune system to fight foreign invaders
- Allow us to see the world in color (instead of black & white)



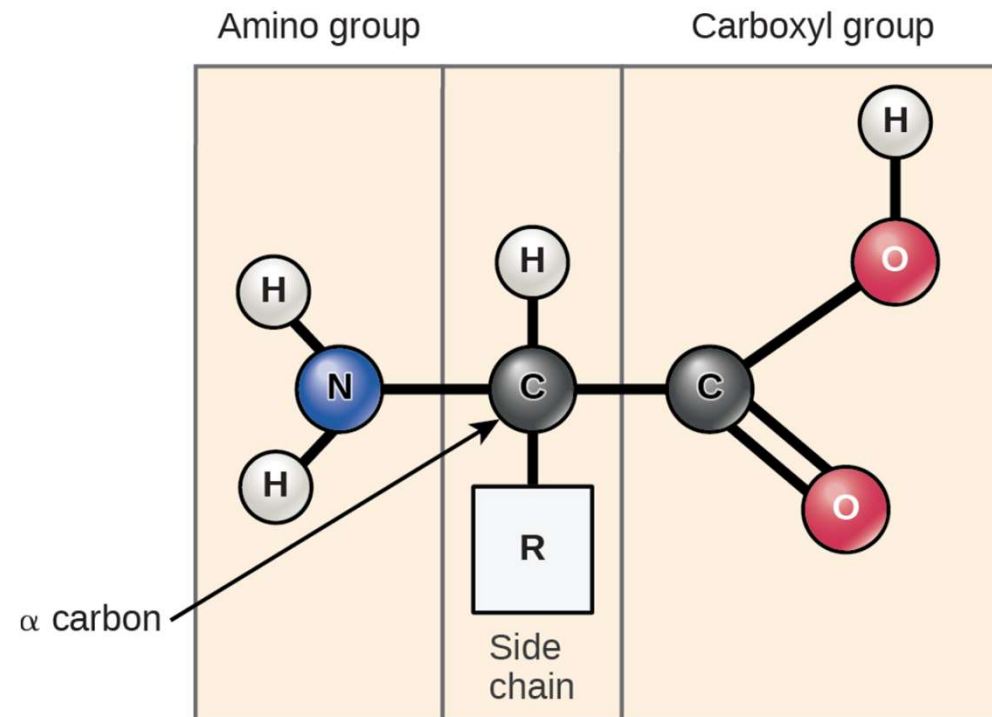
- Even though proteins do so many different things, they are all made the same way out of the same building blocks: amino acids
 - **Amino acids** = monomers of proteins, the basic building blocks
 - Join together to make short **peptides**
 - Long peptides = **polypeptides** = proteins



- There are 20 different amino acids
 - The sequence of amino acids is different for each protein, which is how we can make so many different kinds of proteins
 - Human body can make 11 of 20 from scratch
 - Other 9 must come from our diet: called **essential amino acids**

Essential	Nonessential
Histidine	Alanine
Isoleucine	Arginine
Leucine	Asparagine
Lysine	Aspartate
Methionine	Cysteine
Phenylalanine	Glutamate
Threonine	Glutamine
Tryptophan	Glycine
Valine	Proline
	Serine
	Tyrosine

Amino acids = monomers of proteins, the basic building blocks

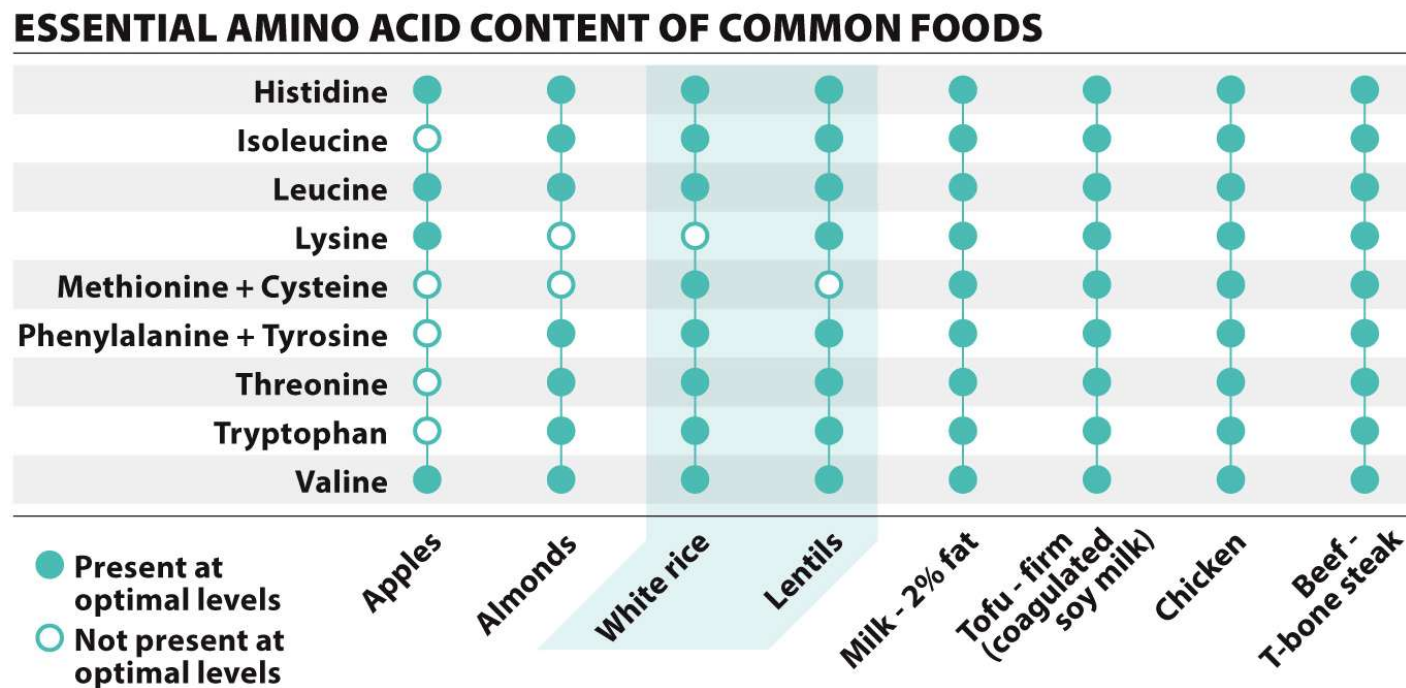


Amino acids with different R side chain

AMINO ACID		
Nonpolar, aliphatic R groups		Glycine
		Alanine
		Valine
		Leucine
		Methionine
		Isoleucine
Polar, uncharged R groups		Proline
		Threonine
		Serine
		Cysteine
		Asparagine
		Glutamine

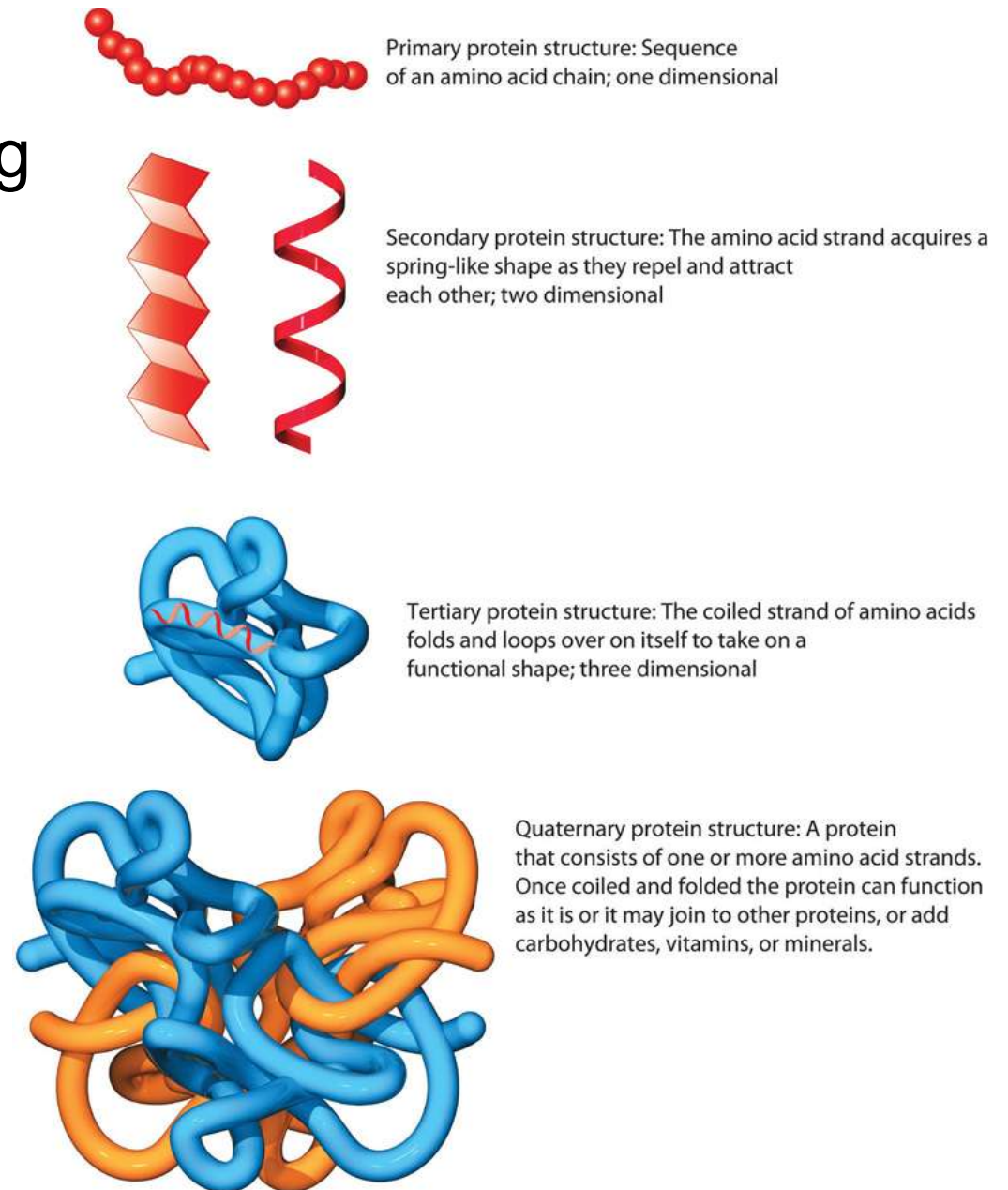
AMINO ACID		
Positively charged R groups		Lysine
		Arginine
		Histidine
Negatively charged R groups		Aspartate
		Glutamate
Nonpolar, aromatic R groups		Phenylalanine
		Tyrosine
		Tryptophan

- Some foods have “complete proteins” with all 9 essential amino acids
 - e.g. meats & dairy, plus plenty of plant sources such as soy & quinoa
 - can combine foods to get all 9: e.g. beans + rice

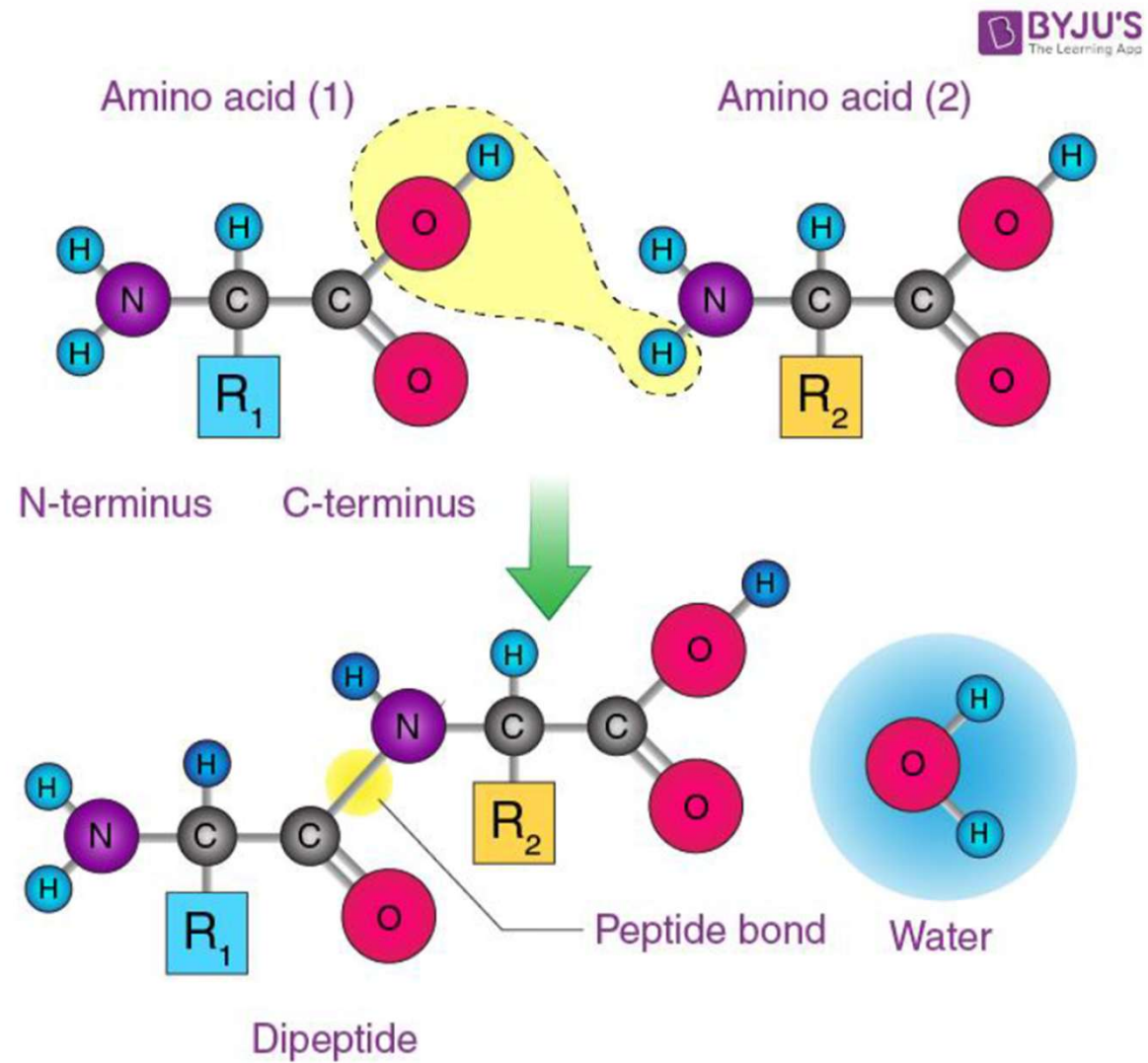


- Amino acids join together to make a long polypeptide chain / protein

– Proteins must be folded into the correct shape/structure before they can function



Peptide Bond



Peptide Bond

- **Characteristics of Peptide Bonds**

- 1. Peptide bonds are strong with partial double bond character:
 - They are not broken by heating or high salt concentration.
 - They can be broken by exposing them to strong acids or bases for a long time at elevated temperature, and also by some specific enzymes (digestive enzymes).
- 2. Peptide bonds are rigid and planar bonds; therefore, they stabilise protein structure.
- 3. Peptide bond contains partial positive charge groups (polar hydrogen atoms of amino groups) and partial negative charge groups (polar oxygen atoms of carboxyl groups).

- **Different Forms of Peptide Bond**

- **Dipeptide** = contains 2 amino acid units.
- **Tripeptide** = contains 3 amino acid units.
- **Tetrapeptide** = contains 4 amino acid units.
- **Oligopeptide** = contains not more than 10 amino acid units.
- **Polypeptide** = contains more than 10 amino acid units, up to 100 residues.
- **Macropeptides** = made up of more than 100 amino acids.

■ Protein structure

Protein's shape is critical to its function

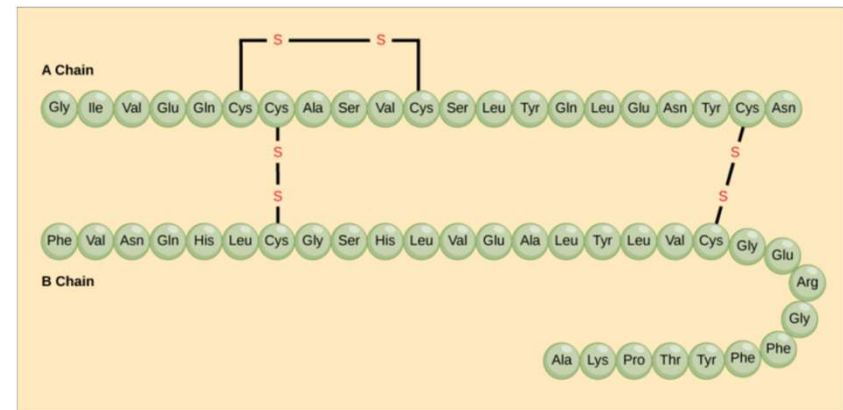
Four levels of structure

■ Primary Structure

- amino acids have just been joined together (with peptide bonds) to make a long polypeptide chain
- *not functional yet*

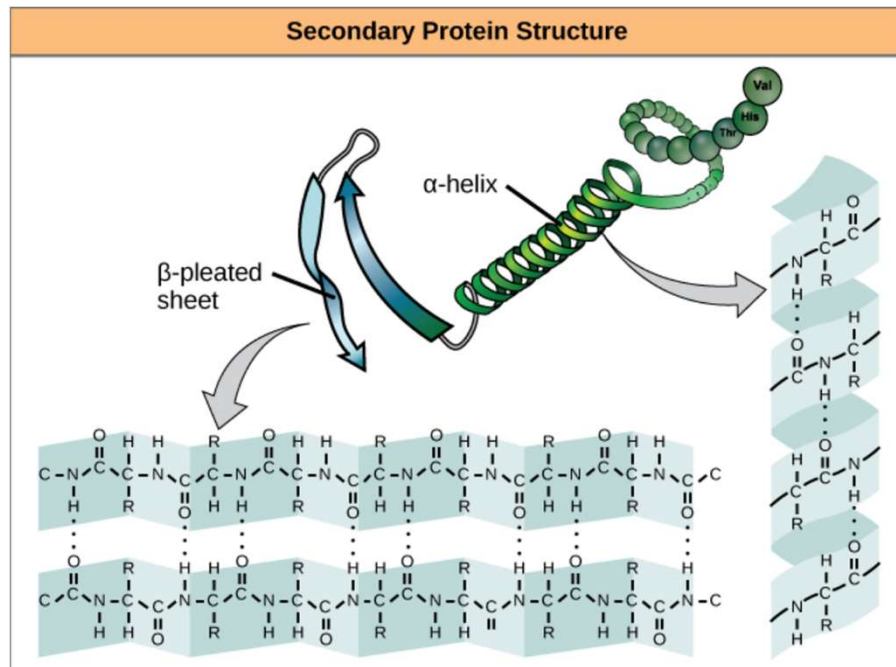
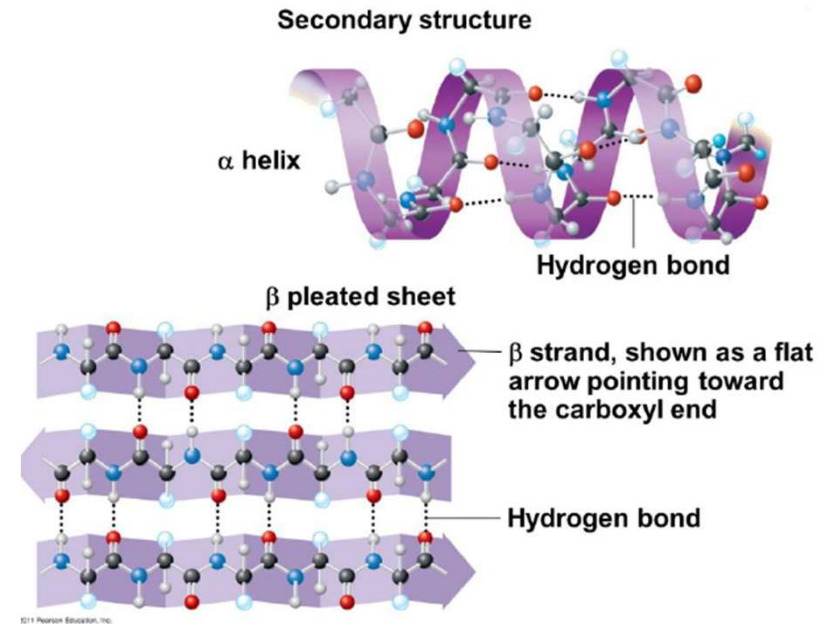


Primary protein structure: Sequence of an amino acid chain; one dimensional



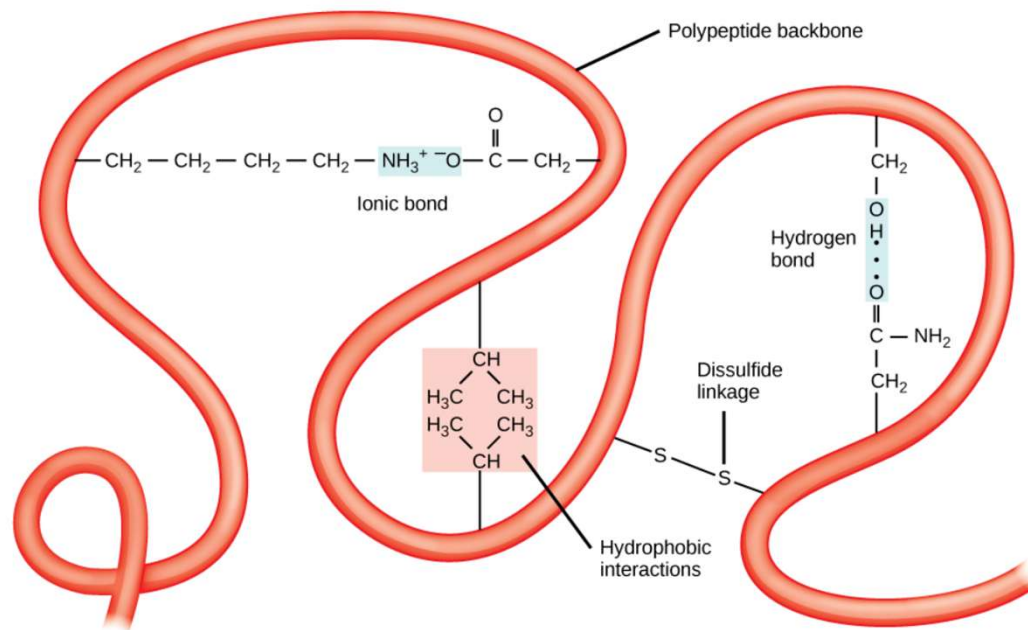
pancreatic hormone insulin

- **Secondary structure:**
the polypeptide chain begins to fold up (through hydrogen bonding)
– *still not functional yet*



The amino acid strand acquire a spring-like shape as they repel and attract each other, 2-d

- **Tertiary structure:** polypeptide chain is now fully folded & ready to do simple work
 - 1st point a protein is functional
 - *e.g. a simple antibody in immune system*



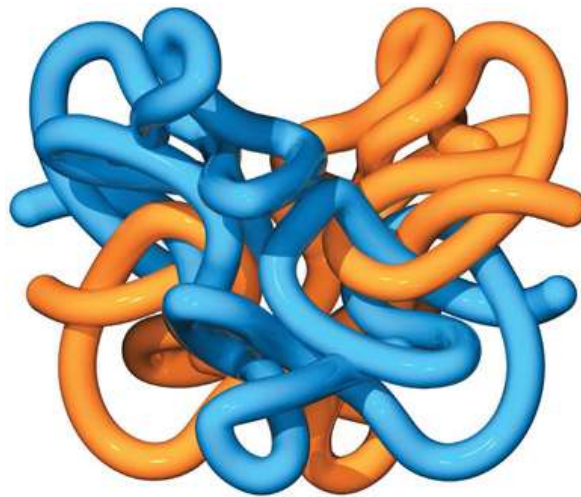
The coiled strand of amino acids folds and loops over on itself to take on a functional shape, 3-d

Fig. 3.29 A variety of chemical interactions determine the proteins' tertiary structure. These include hydrophobic interactions, ionic bonding, hydrogen bonding, and disulfide linkages

■ Quaternary structure:

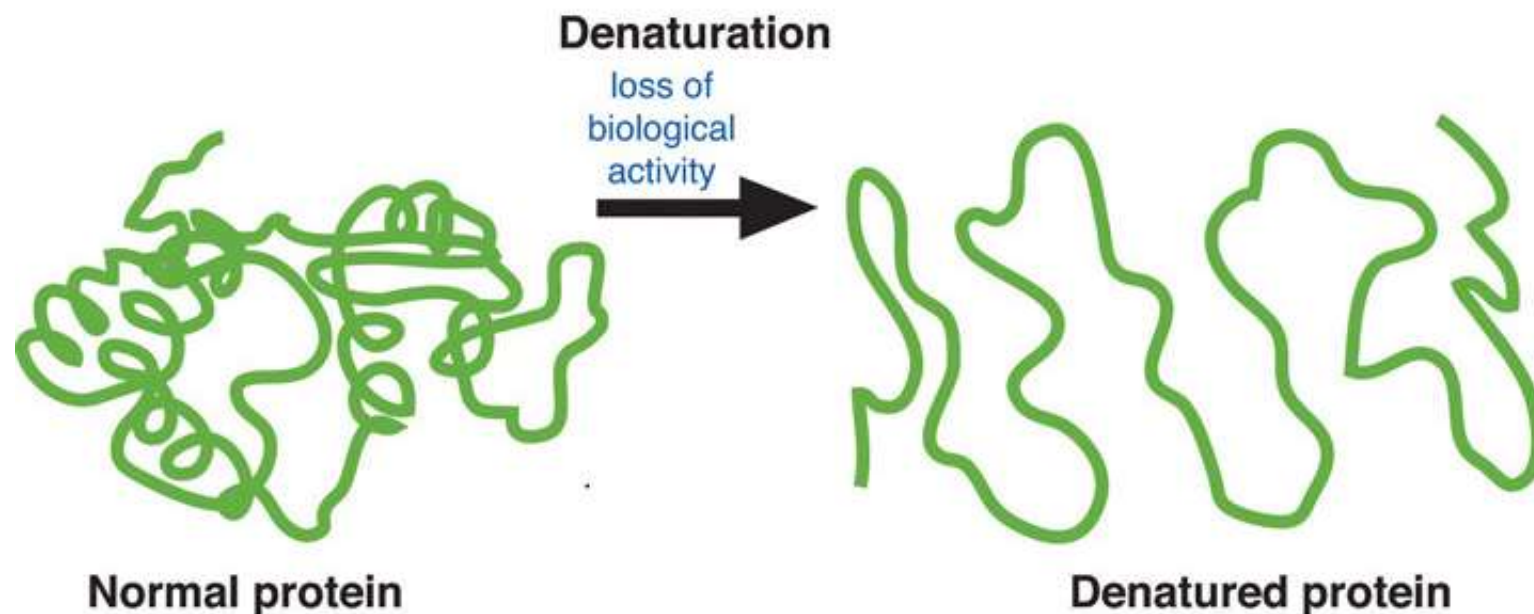
2 or more polypeptide chains bind together to do more complex work

- *e.g. four join together to make hemoglobin, which carries oxygen from lungs all around the body*
- *e.g. dozens join together to make ribosomes, which build new proteins*

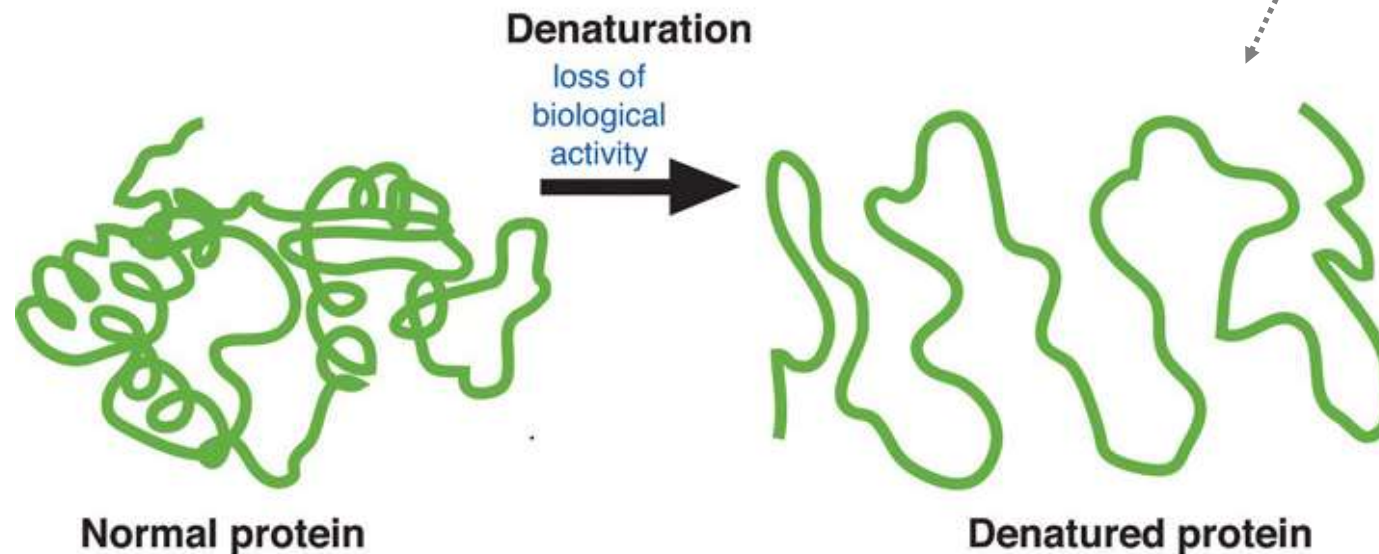


Quaternary protein structure: A protein that consists of one or more amino acid strands. Once coiled and folded the protein can function as it is or it may join to other proteins, or add carbohydrates, vitamins, or minerals.

- If a protein becomes unfolded or mis-folded, it can no longer function
 - **Denaturation** = unfolding of a protein
 - Many things can cause a protein to denature, thereby losing its function permanently



- Remember: proteins do nearly everything to keep us alive & many things can cause proteins to denature
 - If our bodies get too hot, too acidic, too basic, too salty, etc., our proteins will denature & we die
- Remember: living things must maintain homeostasis – now you know why!



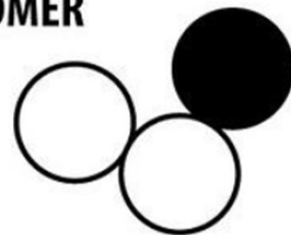
- How does a cell know which *proteins* to make in order to build & maintain our bodies?
 - the information is stored in **nucleic acids**
e.g. deoxyribonucleic acid (DNA)
& ribonucleic acid (RNA)



- How does a cell get the energy out of *carbohydrates* & send it to where it's needed in our bodies?
 - the energy is carried by **nucleic acids**
 - e.g. ATP

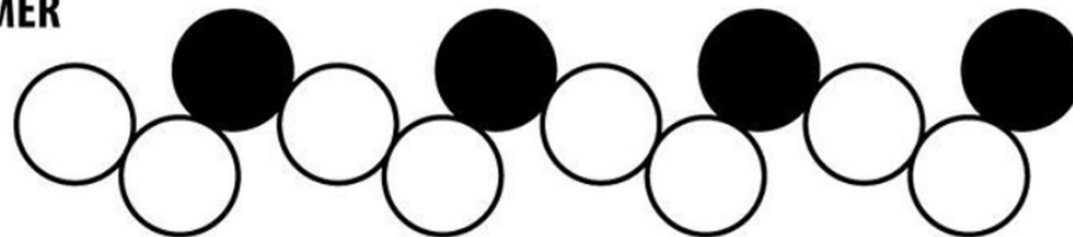
- **Nucleic acids** are polymers made up of chains of nucleotides
 - **Nucleotides** = monomers, each of which has at least 1 phosphate, a sugar, & a base

MONOMER



A monomer is a small molecule.

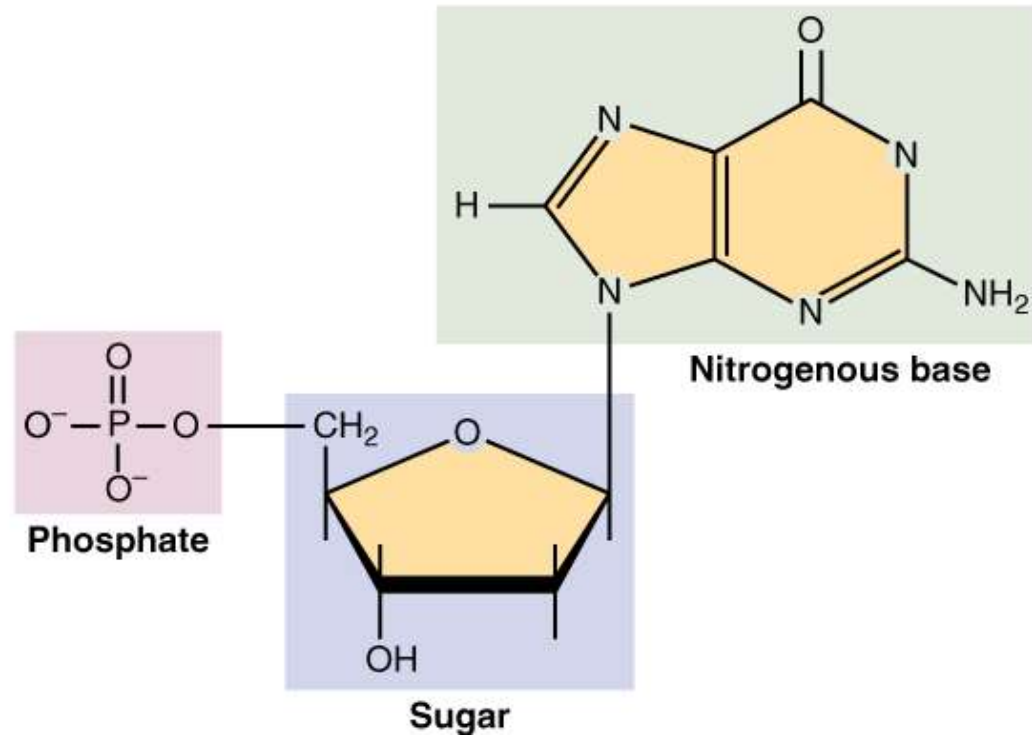
POLYMER



A polymer is a long-chain molecule made up of a repeated pattern of monomers.

■ Nucleic Acids

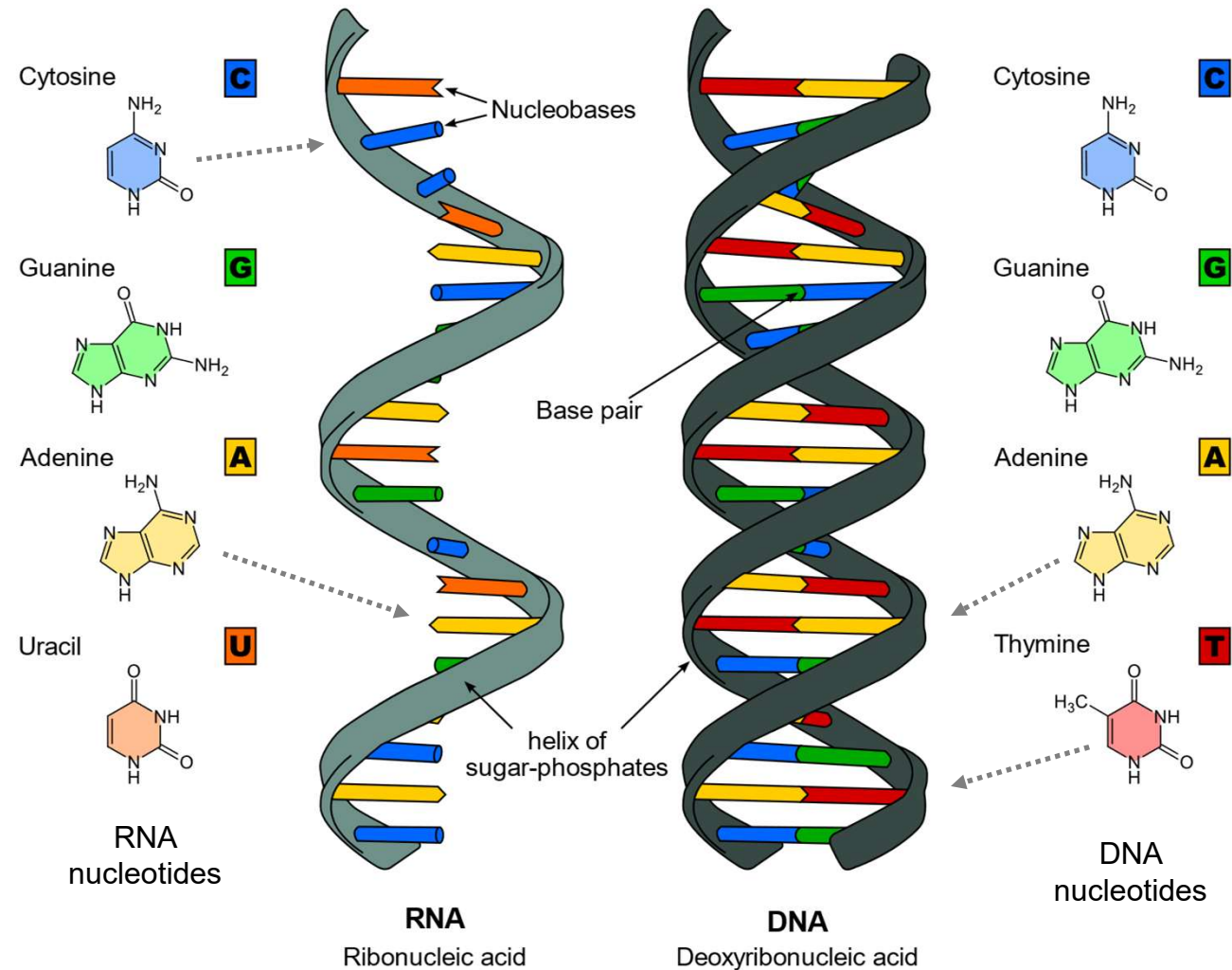
one nucleotide monomer has 3 components



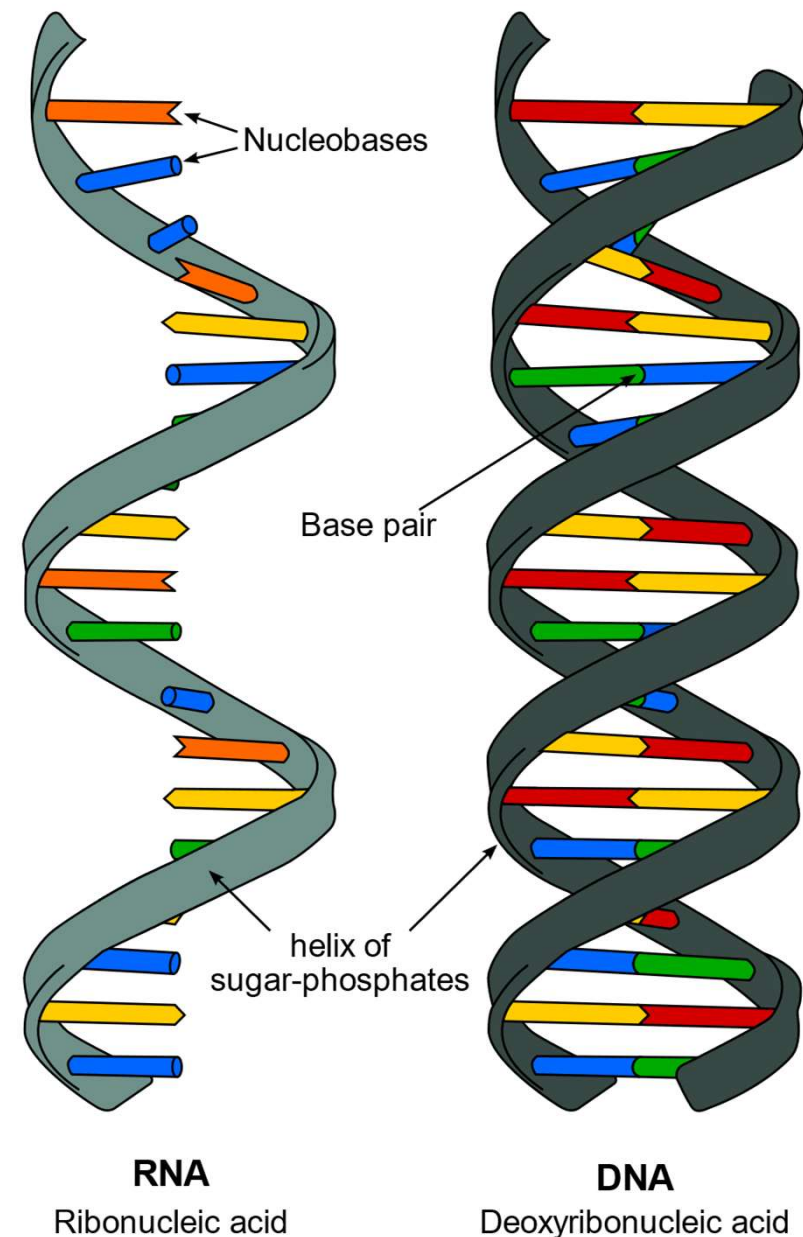
- A chain of nucleotides make up a nucleic acid
 - e.g. **DNA & RNA**

DNA contains one of four possible nitrogenous bases: adenine (A), guanine (G), cytosine (C), and thymine (T).

RNA contains A, U (Uracil), G, and C.

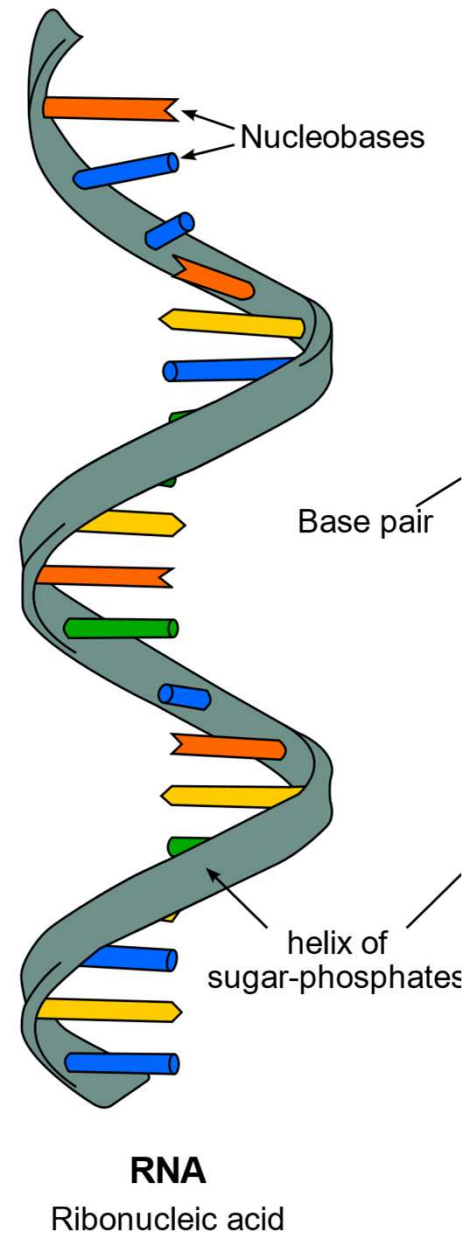


- **DNA:** locked in the nucleus of cells, stores our genetic information
- **RNA:** acts as a messenger, carrying instructions from DNA out to the rest of the cell



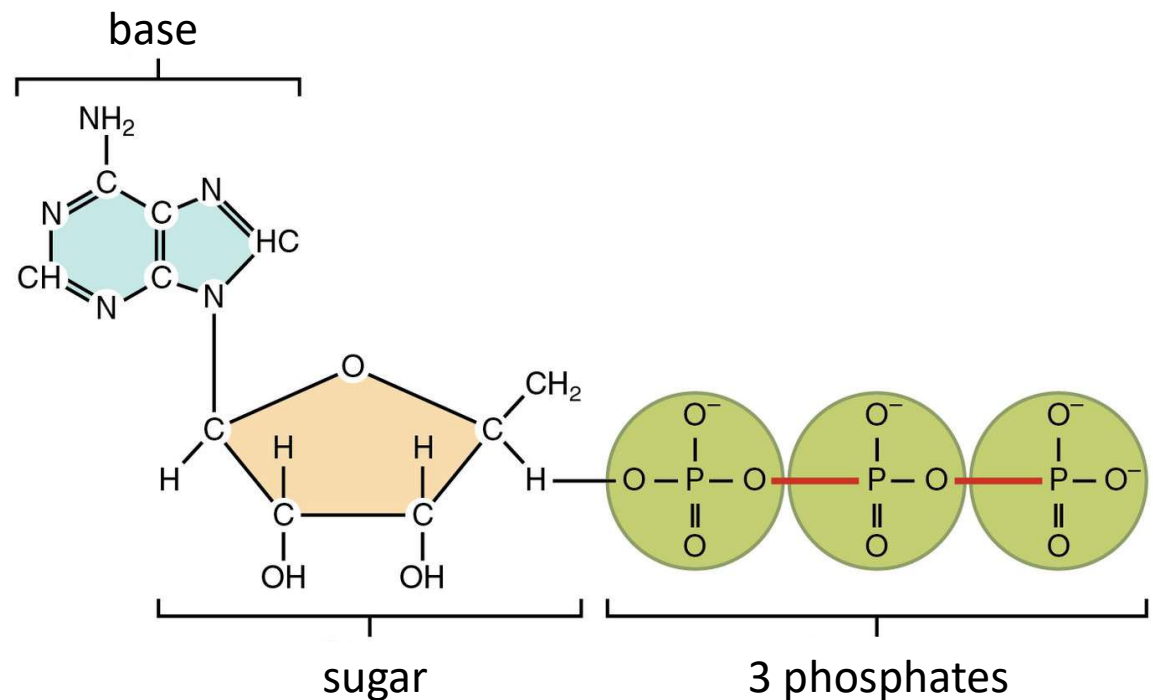
■ 4 main types of RNA

1. messenger RNA (mRNA)
2. ribosomal RNA (rRNA)
3. transfer RNA (tRNA)
4. microRNA (miRNA)



- Some nucleotides stay as monomers permanently (never join together to make a nucleic acid)
 - e.g. **ATP**: carries energy (from breaking down carbohydrates) around the cell to where it's needed

ATP = adenosine triphosphate: 3 phosphates help the molecule carry energy



- **Lipids** = all are hydrophobic molecules
 - *Made by proteins, which are directed by nucleic acids, using the energy from carbohydrates*
- Many uses, including energy storage, waterproofing, hormones, cell membranes, etc.
 - have diverse chemical structures, with no defined monomers or polymers
- Can break them up into 4 main groups:

1. Fats

- long-term energy storage & insulation

2. Waxes

- waterproof molecules

3. Sterols

- hormones, support cell membranes, etc.

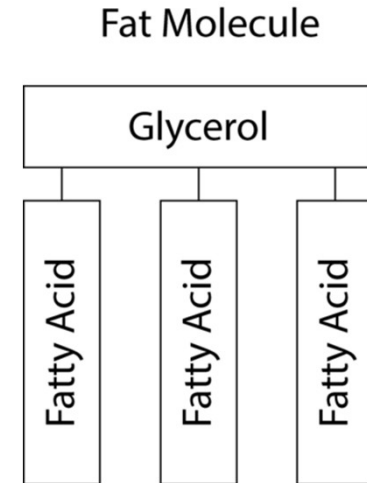
4. Phospholipids

- form cellular membranes

- Lipid group 1: **Fats**

- Fats (& oils too) are molecules called triglycerides

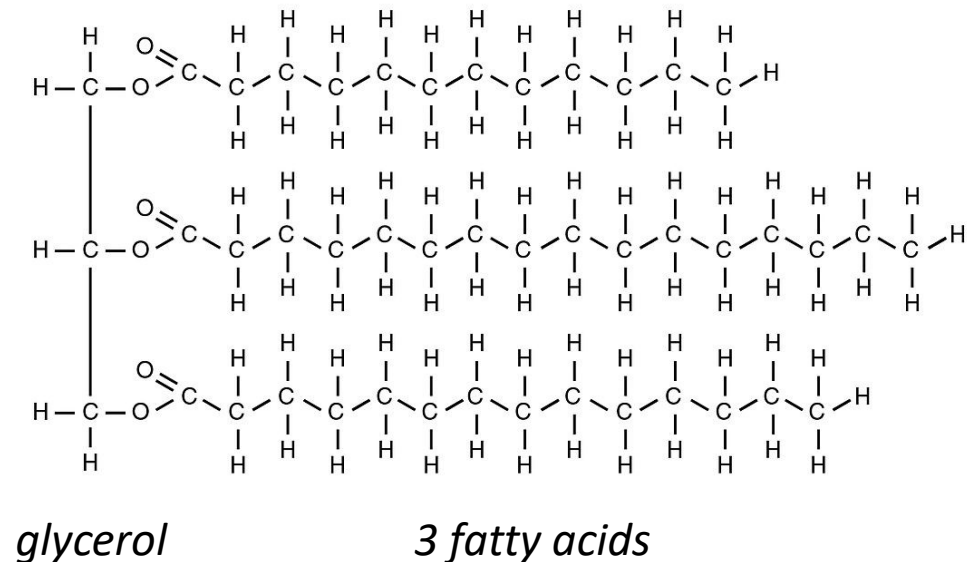
- **triglyceride** = 3 fatty acids attached to a glycerol



- **fatty acids** = long chains of C & H (with some O)

- *will be important to know to distinguish between saturated & unsaturated fats*

(a) Triglyceride



- Fats & oils are used for energy storage & insulation

- One molecule of fat stores twice as much energy as one molecule of carbohydrates

- Fat also provides insulation



Because fats store such large amounts of energy, a strong taste preference for fats over other energy sources has evolved in animals.

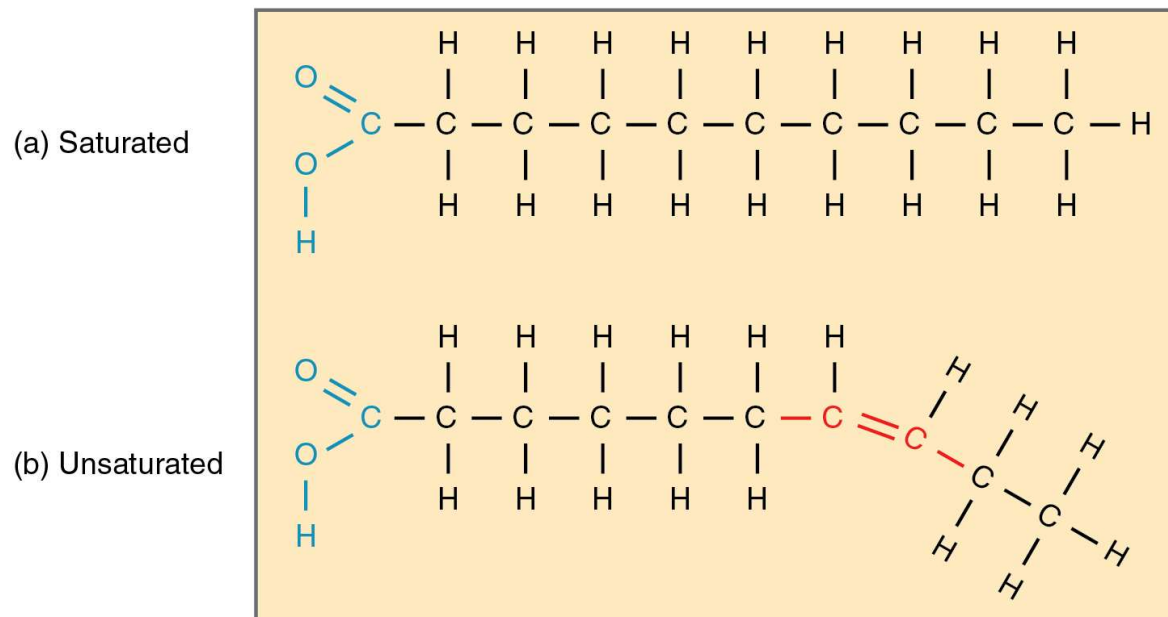


The Emperor penguin benefits from the insulating effect of fat and is able to thrive in the extreme cold of Antarctica.

Figure: <https://pixabay.com/photos/hamburger-p-french-fries-belly-2683042/>

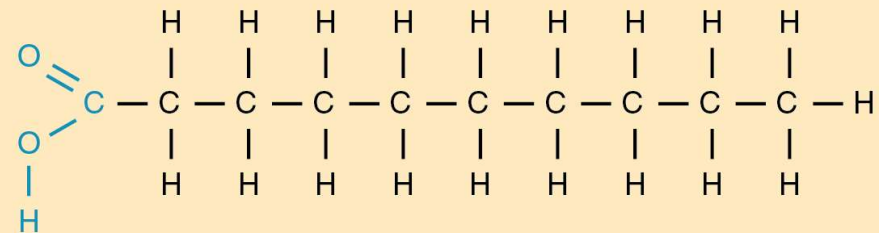
Figure: https://commons.wikimedia.org/wiki/File:Emperor_Penguin_Manchot_empereur.jpg

- Fats & oils can be saturated or unsaturated
 - Fatty acids that have carbon chains “saturated” with hydrogen (completely filled) = **saturated fats**
 - They are straight, so they line up well & stack tightly = solid at room temperature

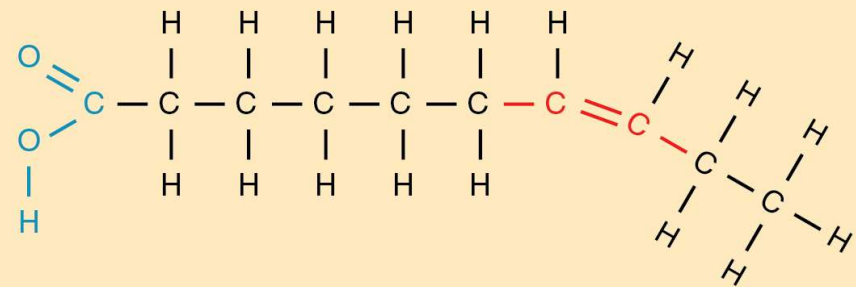


- Fats & oils can be saturated or unsaturated
 - Fatty acids that have carbon chains missing some hydrogen (not saturated) = **unsaturated fats**
 - Don't line up or stack well together = liquid at room temperature
 - Usually call these **oils**

(a) Saturated



(b) Unsaturated



- Fats & oils can be saturated or unsaturated

- Animals tend to make saturated fats

- e.g. beef fat & butter



- Plants tend to make unsaturated fats

- e.g. vegetable, corn, & olive oils



Figure: <https://www.flickr.com/photos/30478819@N08/48612236702>

Figure: <https://www.pxfuel.com/en/free-photo-odsys>

- Unsaturated fats can become “**hydrogenated**” or “partially hydrogenated”
 - Hydrogenation = process that adds hydrogen to unsaturated fats to make them more saturated
 - Partially hydrogenated = some have been made saturated, but not all
 - Hydrogenated fats don’t exist in nature
- So why hydrogenate?
 - Store better
 - Preserve flavor longer
 - Cheaper than saturated fats

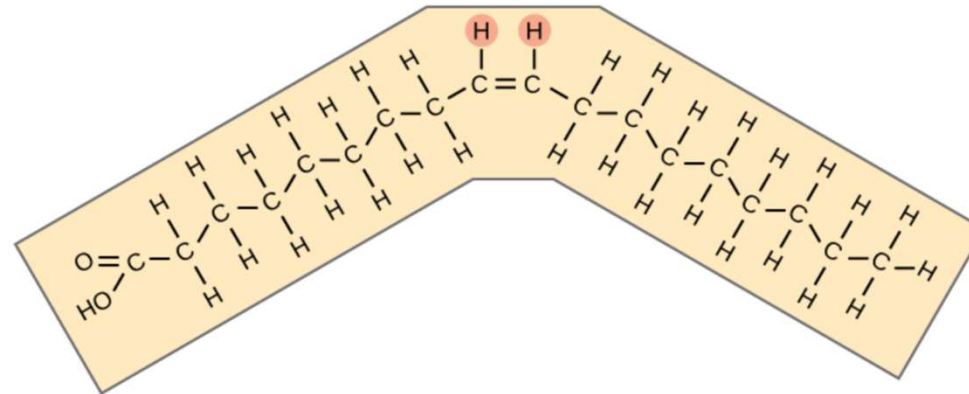


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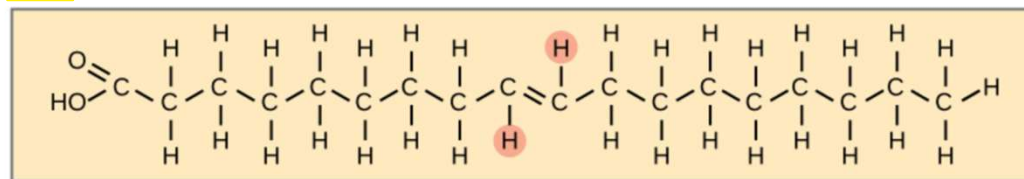
- Trans fats are made during the hydrogenation process
 - **Trans fats** = unsaturated fats that have been hydrogenated AND have had their molecular confirmation altered

Unsaturated fatty acids

Cis oleic acid



Trans oleic acid



- Different fats are processed differently in the body
 - Unsaturated fats (that are not hydrogenated) are *generally* healthier (in moderation) & have been linked to:
 - Lower bad cholesterol
 - Reduced heart disease
 - Lower blood pressure



Figure: <https://www.pxfuel.com/en/free-photo-odsys>

- Different fats are processed differently in the body
 - Hydrogenated & trans fats are VERY unhealthy & linked to:
 - Rise in bad cholesterol
 - Heart disease
 - Diabetes & other metabolic disorders
 - Unfortunately, hydrogenated & trans fats are now incredibly common in our food supply
 - Found in fried foods, commercial baked goods, & highly processed foods



- Different fats are processed differently in the body

- Saturated fats are *generally* unhealthy & linked to:

- Colon, breast, ovarian, & other cancers
 - Rise in bad cholesterol
 - Heart disease
 - *See next slide for the saturated fats that might be an exception*



Figure: <https://www.flickr.com/photos/30478819@N08/48612236702>

Figure: <https://pixabay.com/photos/high-fat-foods-pastries-cheeses-1487599/>

- There are several different types of saturated fats
 - Foods with saturated fat have some combination of the different types, each in different amounts
 - Most types of saturated fats are generally unhealthy, but some are generally healthy



- ◆ e.g. foods with higher levels of palmitic or myristic saturated fat = unhealthy
 - dairy, meat, & eggs



- ◆ e.g. foods with higher levels of lauric or stearic saturated fat = healthy
 - coconut & dark chocolate

- Lipid group 2: **waxes**
 - **Waxes** = fatty acids combined with other molecules, forming a stiff, waterproof material
 - Make up stiff honeycombs
 - Keep fur, feathers, leaves, & fruits waterproof

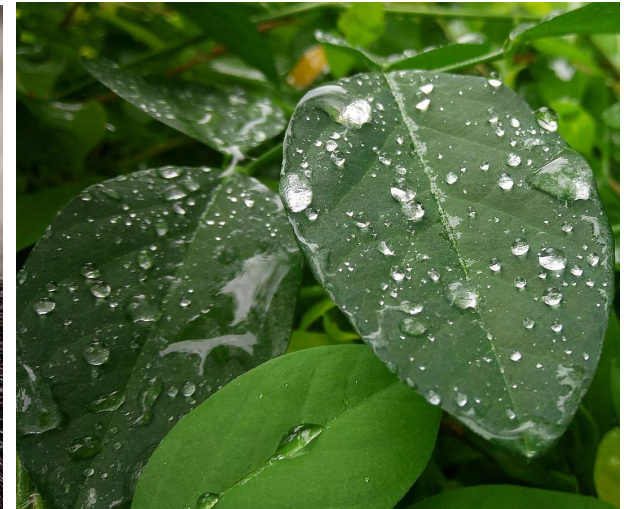


Figure: https://commons.wikimedia.org/wiki/File:Western_honey_bee_on_a_honeycomb.jpg

Figure: <https://www.hippopx.com/en/otter-wet-lutra-lutra-zoo-fur-marten-154984>

Figure: https://commons.wikimedia.org/wiki/File:Wet_Leaves.jpg

■ Lipid group 3: **sterols**

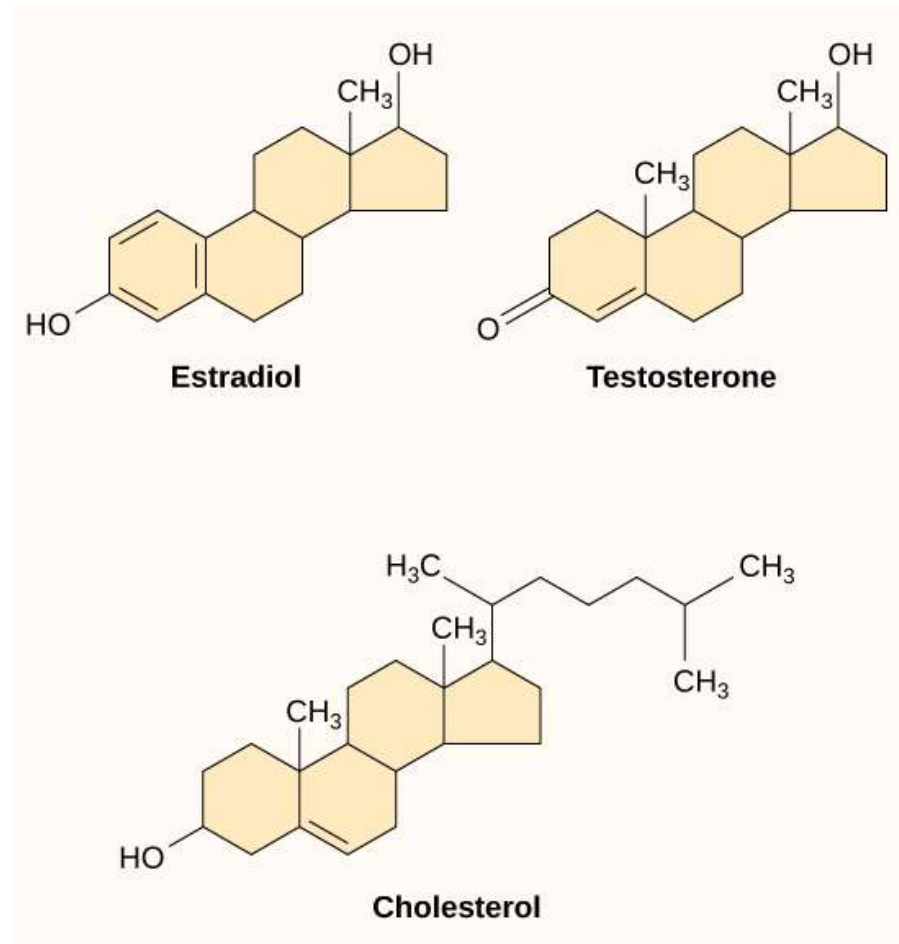
- Includes cholesterol & hormones such as estrogen & testosterone

Steroid hormones

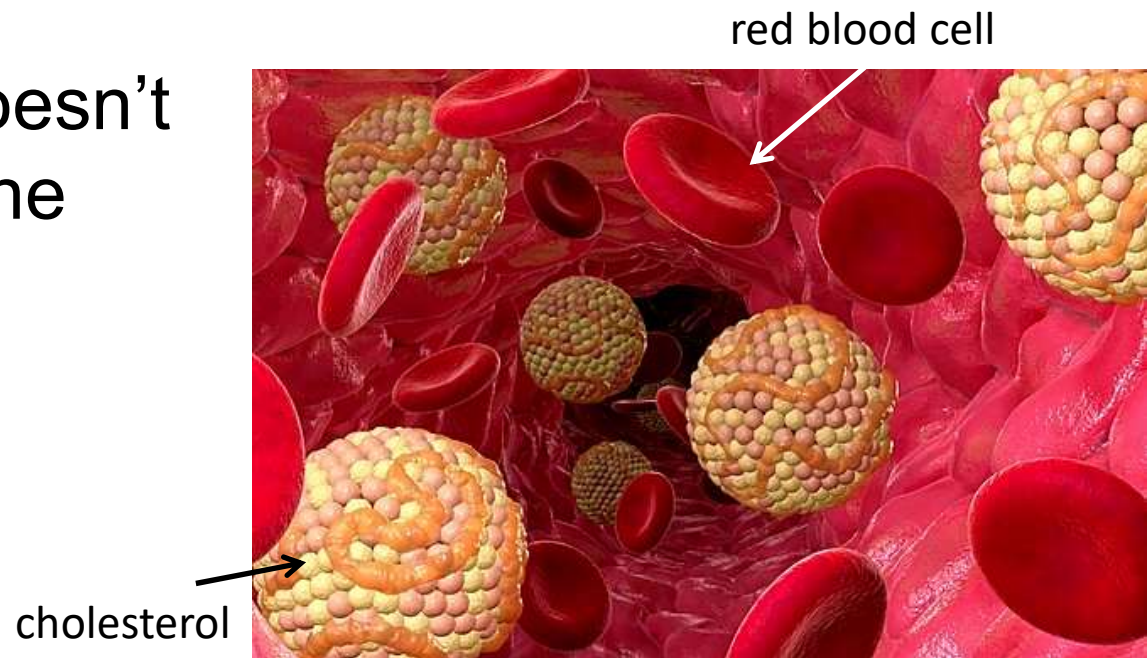
- *Examples = testosterone and estrogen*
- *Regulate sexual development, maturation, sex cell production, mood, and body growth*

Cholesterol

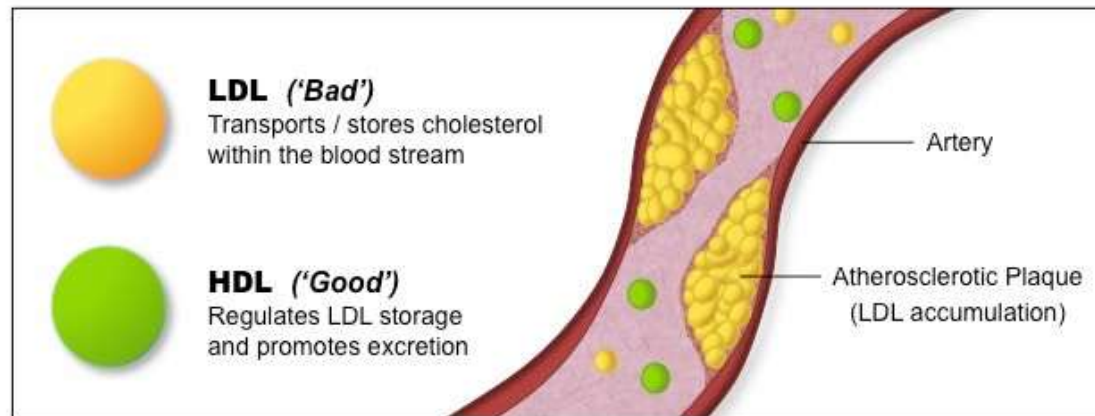
- *Important component of cell membranes in animals*
- *Can attach to and thicken vessel walls and may cause serious health problems*



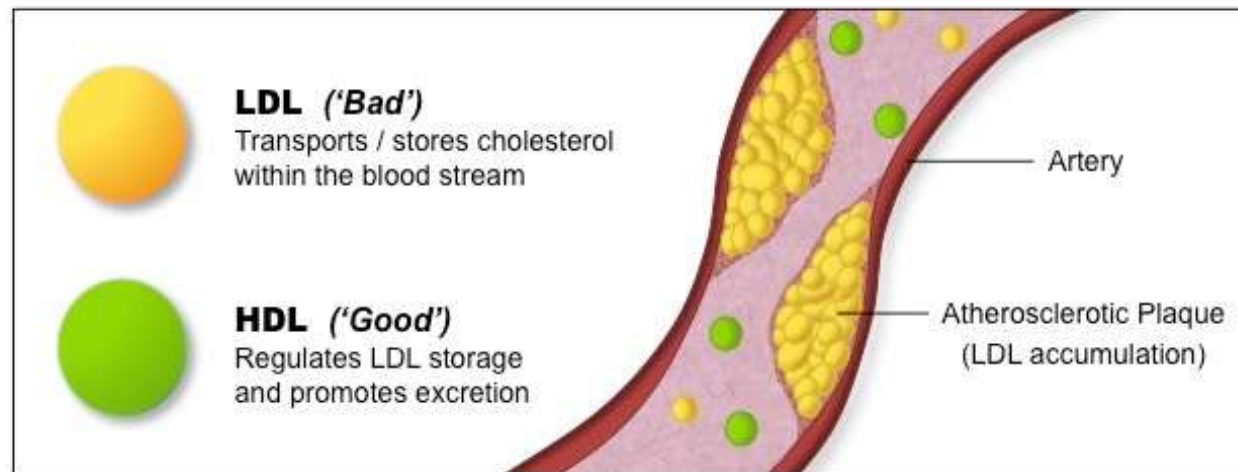
- **Cholesterol** = necessary for cell health
 - Animals make their own – no need to eat any
 - Remember: lipids like cholesterol are hydrophobic, which means they clump up in water – blood is mostly water
 - Cholesterol doesn't travel well in the blood stream (clumps up)



- Proteins bind on to cholesterol to carry it through the blood stream = **lipoproteins**
 - 2 types of lipoproteins
 - **HDL** (high density lipoproteins)
 - Carry cholesterol to liver → processed and removed by the body
 - Often referred to as “good cholesterol”



- 2 types of lipoproteins
 - **LDL** (low density lipoproteins)
 - Keep cholesterol inside bloodstream → deposited on artery walls
 - Increases the chance of heart attack and stroke – “bad cholesterol”



■ Cholesterol, lipoproteins, & health

– These things **raise LDL** (bad) levels:

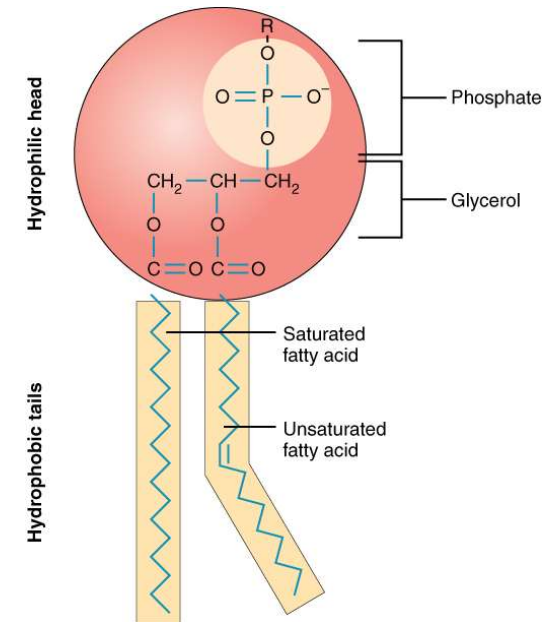
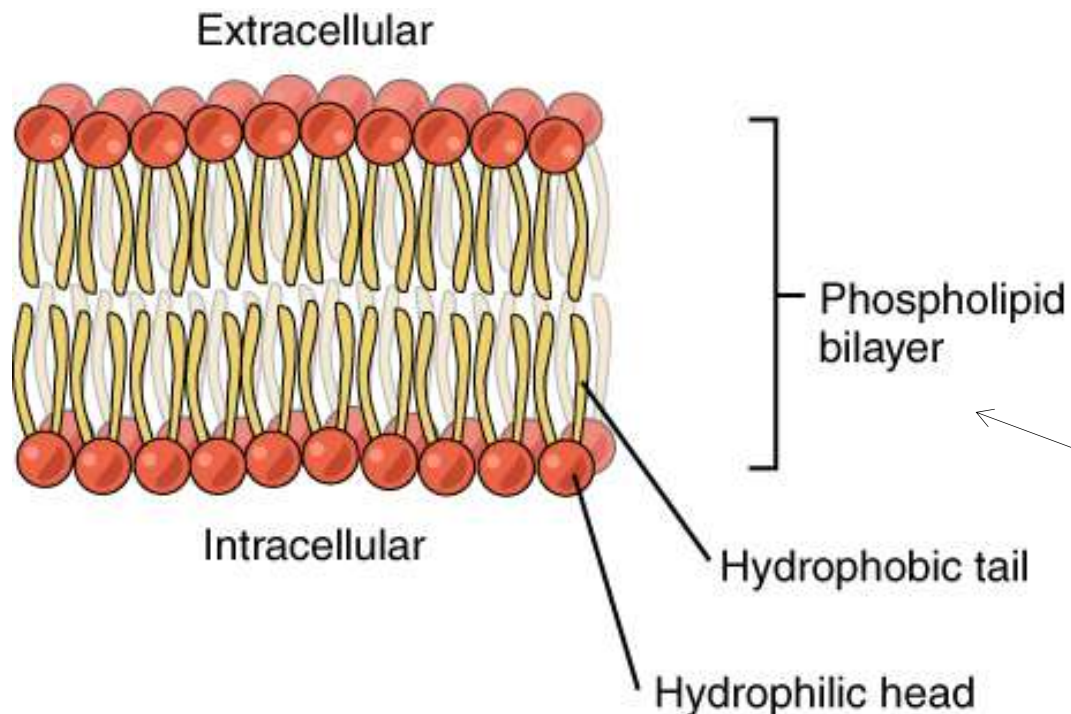
- Too much saturated, hydrogenated, & trans fat in diet
- Too much cholesterol in diet (comes from meat, eggs, & dairy)
- Tobacco use (through lungs, skin, or ingestion)

– These things **raise HDL** (good) levels:

- Exercise – *activity causes HDL production*
- Eating healthy (good fats, low/no cholesterol)
- Avoiding tobacco

- Lipid group 4: **phospholipids**

- Make cell membranes
- *Will cover extensively soon*



one
phospholipid

many phospholipids
gather to create cell
membranes

- Animals, plants, mushrooms (fungi), & bacteria seem to be very different
- However, they all use & are built of the same basic organic molecules (carbohydrates, proteins, nucleic acids, & lipids)
- All are also made up of cells
- Up next: *How are cells built & how do they use those organic molecules to keep us alive?*

